Practical 0

Introduction to Android, Introduction to Android Studio IDE, Application Fundamentals

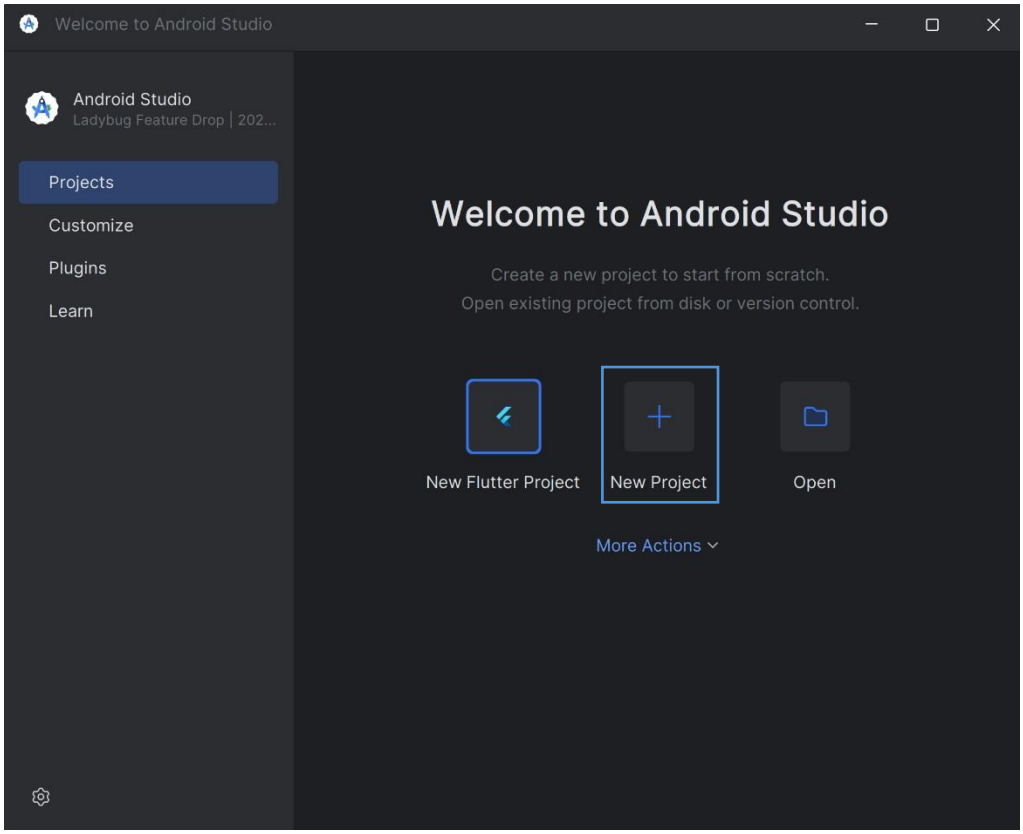
Aim: Creating a simple project with "Hello World" program.

Files Used:

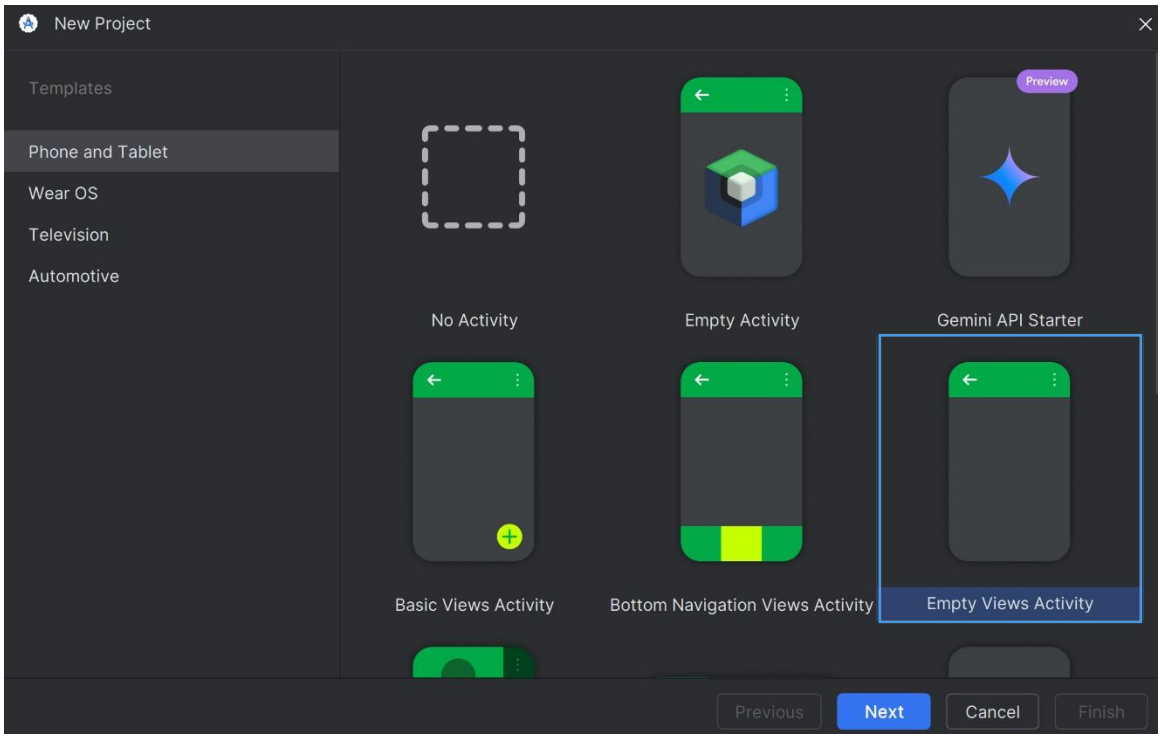
Project Name: Prac_0
Java File: MainActivity.java
XML File: activity_main.xml

Steps to create Project:

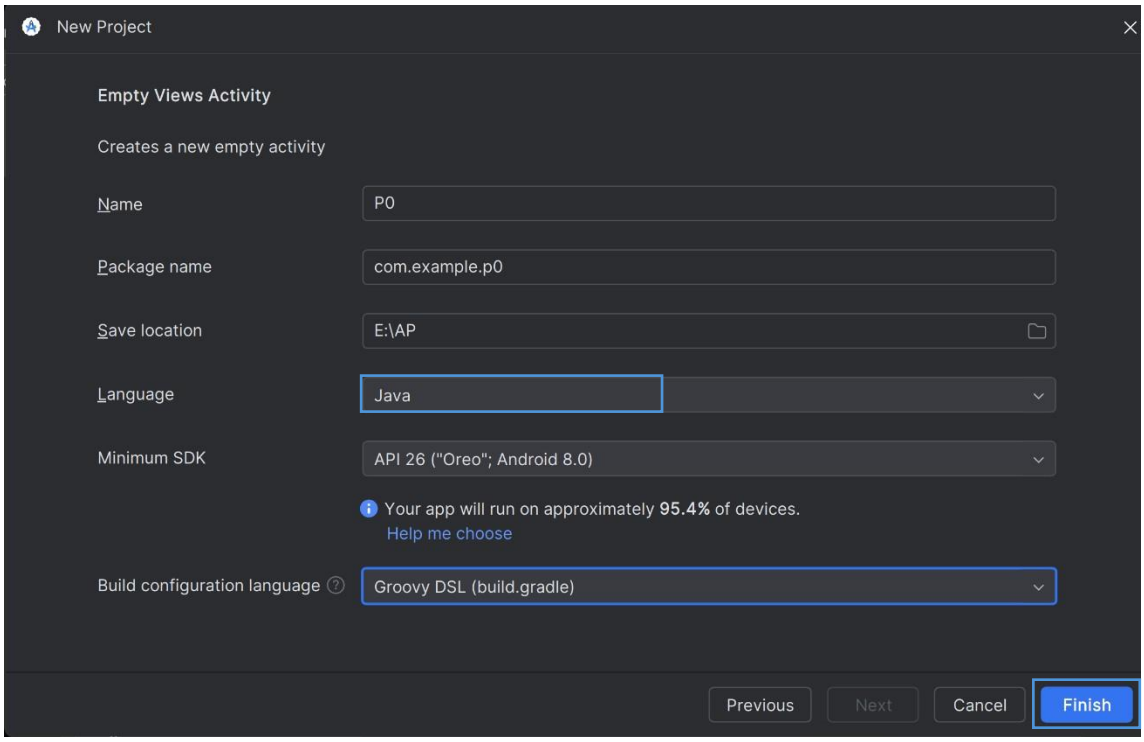
1. Open Android Studio → Click on 'New Project'.



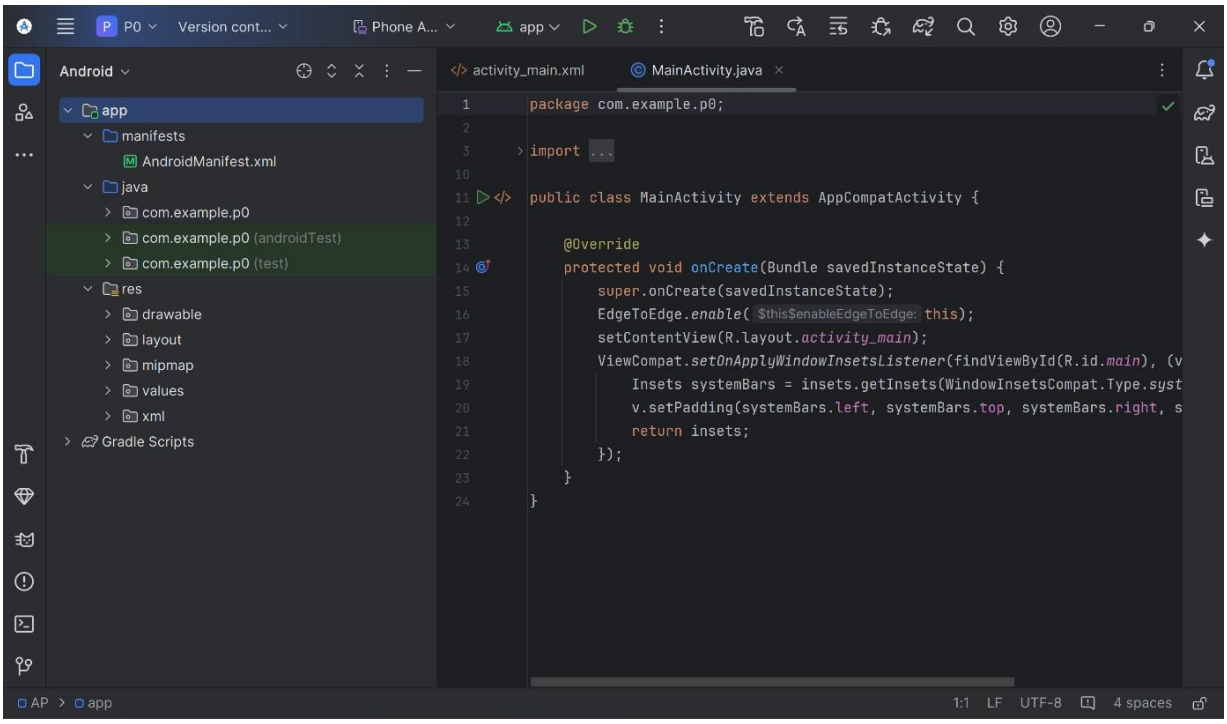
2. Select ‘Empty View Activity’.

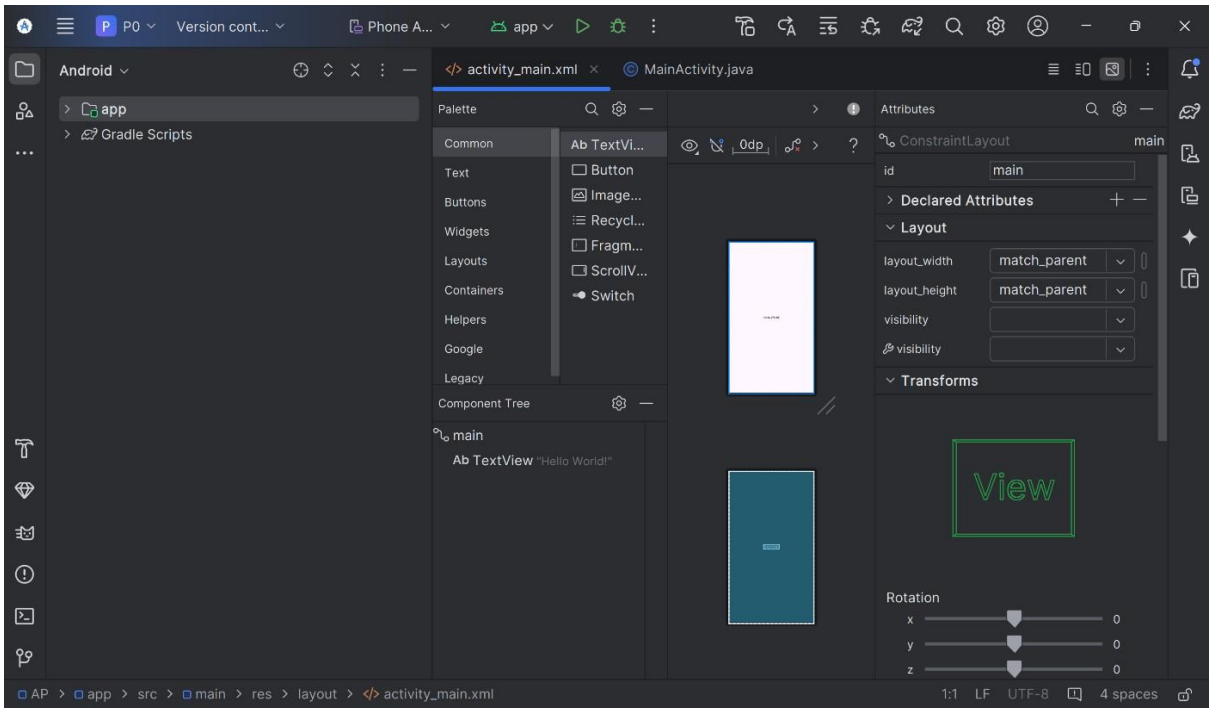


3. Give Project Name as ‘**<u>P0</u>**’ →Select ‘Java’ in language. Click ‘Finish’.



4. ‘Project (P0)’ is successfully created.





Code: MainActivity.java

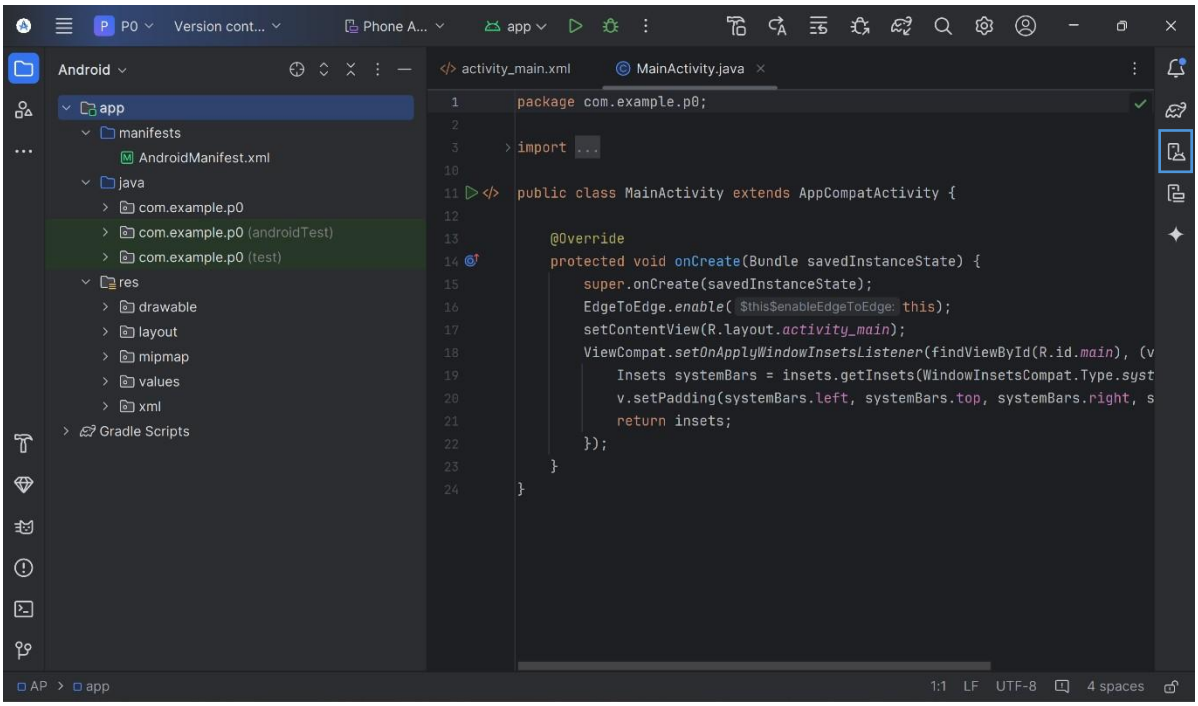
```
package com.example.prac_0;
import android.os.Bundle;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {
   @Override
   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      EdgeToEdge.enable(this);
      setContentView(R.layout.activity_main);
      ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) ->
{
         Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
         v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
         return insets; }); }}
```

activity_main.xml

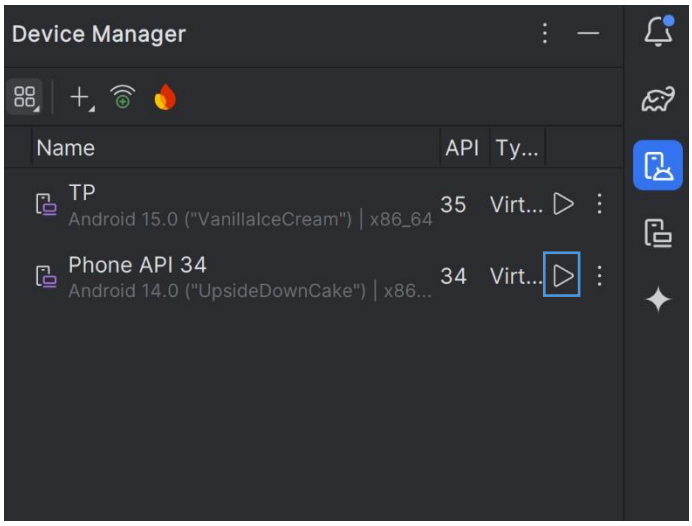
```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:app="http://schemas.android.com/apk/res-auto"
   xmlns:tools="http://schemas.android.com/tools"
   android:id="@+id/main"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   tools:context=".MainActivity">
   <TextView
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Hello World!"
      app:layout_constraintBottom_toBottomOf="parent"
      app:layout_constraintEnd_toEndOf="parent"
      app:layout_constraintStart_toStartOf="parent"
      app:layout_constraintTop_toTopOf="parent" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

Steps to run the project:

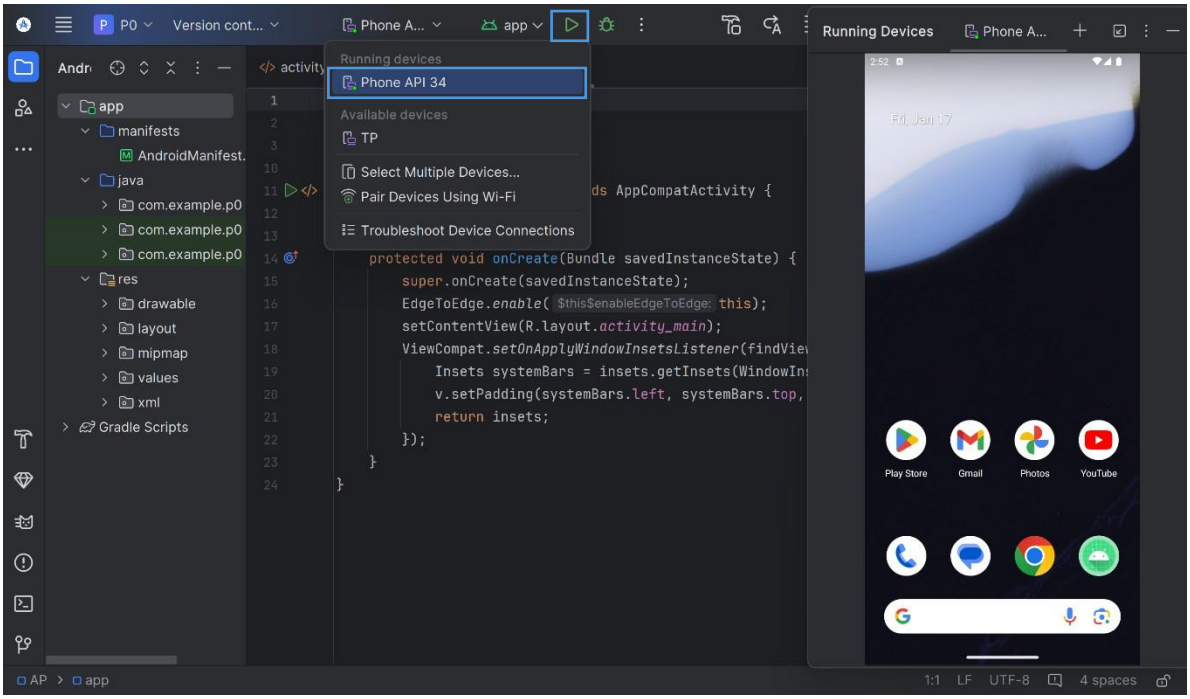
1. Open ‘Device Manager’



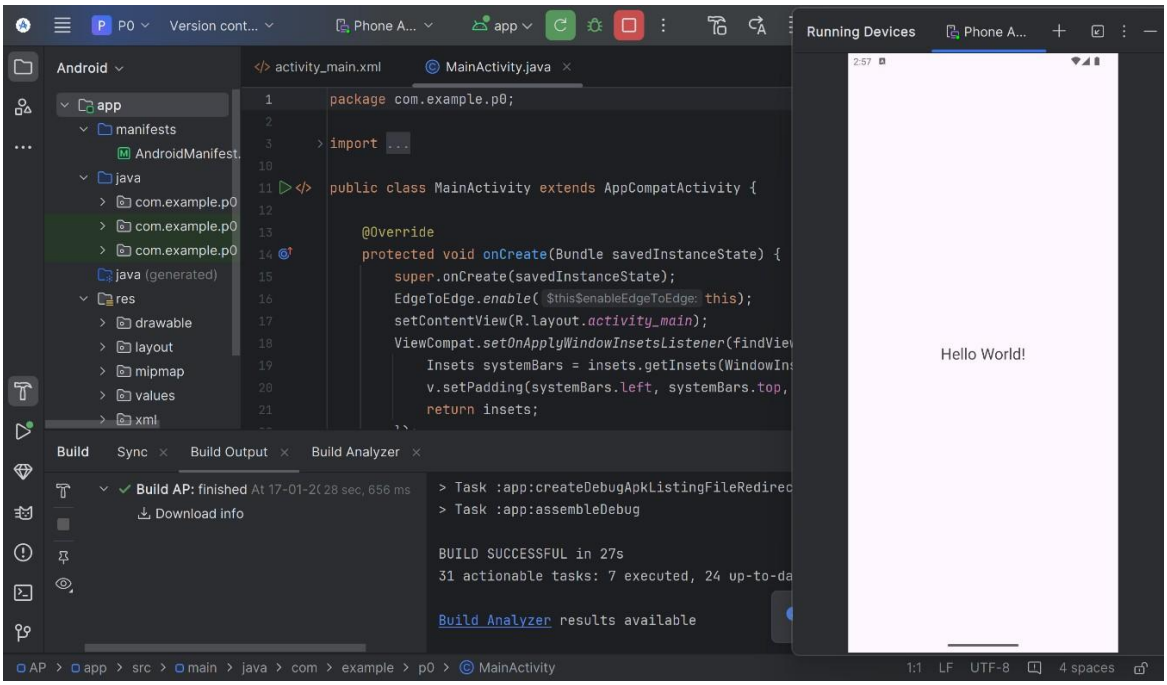
2. Click on ‘Play’ button (which will turn on the device).



3. Select the 'Device (Phone API 34)' → Click on 'Play' button.



Output:



Practical 1

Programming & Android Resources

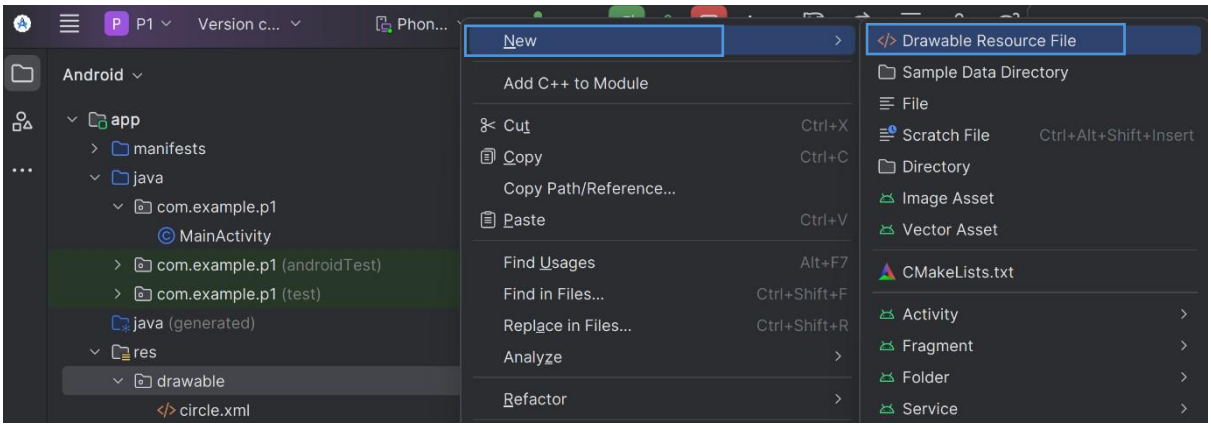
Aim: Use Color, Theme, String, Drawable, Dimension, Image.

Files Used:

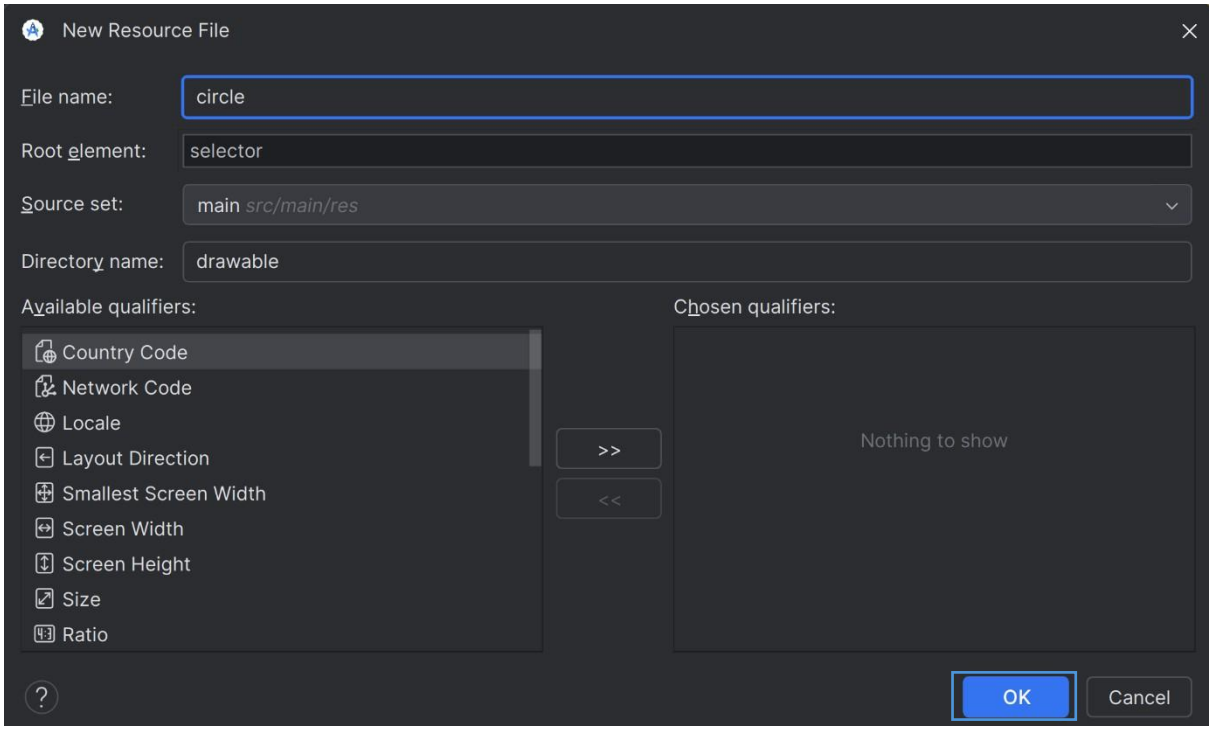
Project Name: **Prac_1**
Java File: **MainActivity.java**
XML File: **activity_main.xml**
circle.xml
random.xml
dimens.xml
themes.xml

Steps to make a Drawable File:

1. Right click on 'Drawable' folder → New → Select 'Drawable Resource File'.

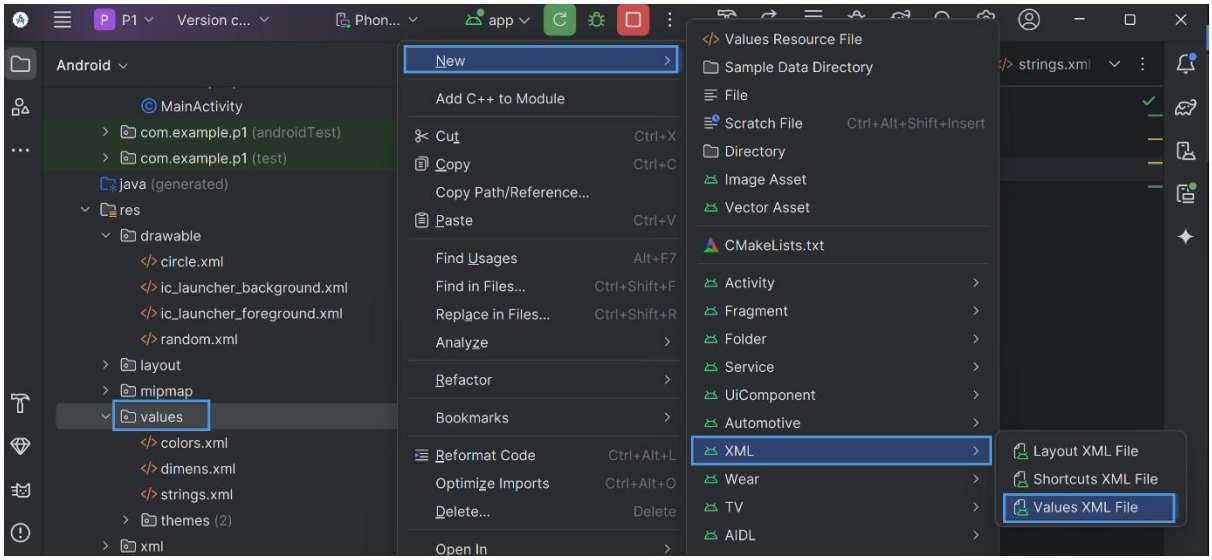


2. Give name as ‘circle’ → Ok.

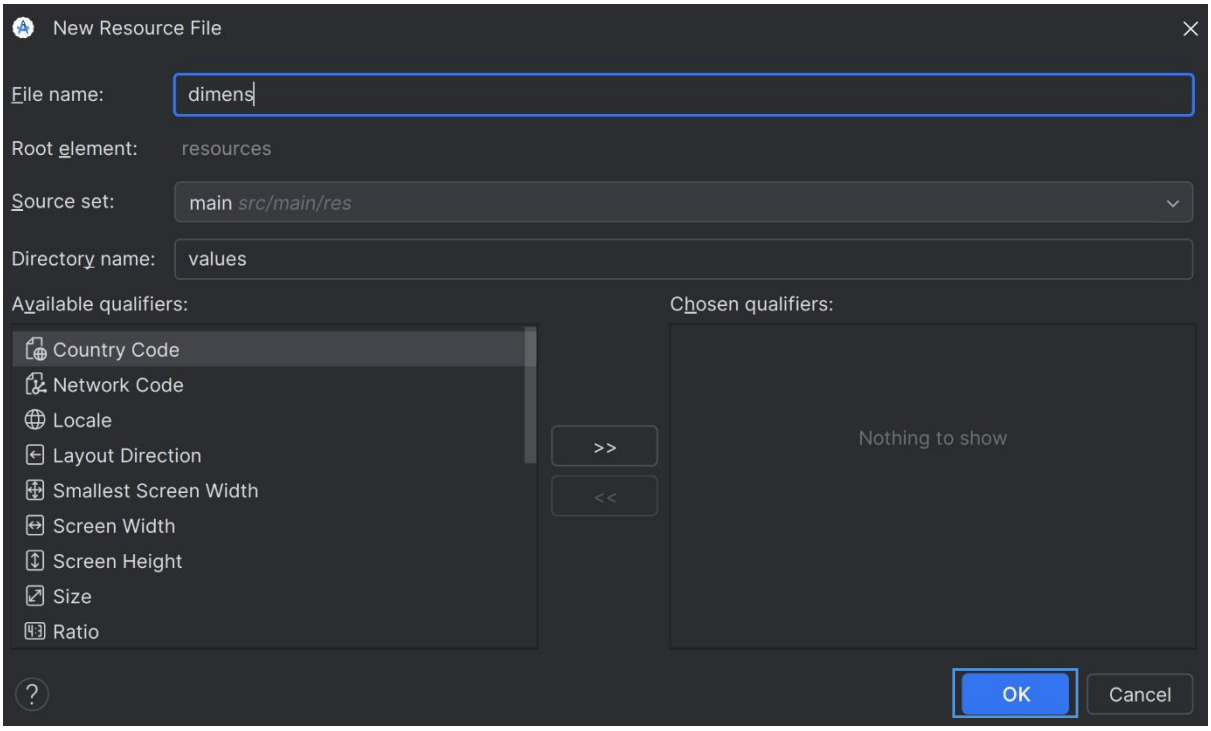


Steps to make a dimens.xml file:

1. Right click on ‘Values’ folder → New → Select ‘XML’ → ‘values XML File’.



2. Give name as ‘dimens’ → Ok.



Code: <u>dimens.xml</u>

```
<resources>
    <dimen name="d1">350px</dimen>
    <dimen name="d2">350px</dimen>
</resources>
```

<u>strings.xml</u>

```
<resources>
    <string name="app_name">pract1</string>
    <string name="t1">This Application will help You.</string>
</resources>
```

colors.xml

```
<resources>
    <color name="black">#FF000000</color>
    <color name="white">#FFFFFFFF</color>
    <color name="d1">#dd6776</color>
    <color name="d2">#ddaacc</color>
    <color name="d3">#ff5566</color>
    <color name="d4">#aa11aa</color>
</resources>
```

themes.xml

```
<resources xmlns:tools="http://schemas.android.com/tools">
    <!-- Base application theme. -->
    <style name="Base.Theme.P1" parent="Theme.Material3.DayNight.NoActionBar">
        <!-- Customize your light theme here. -->
        <!-- <item name="colorPrimary">@color/my_light_primary</item> -->
    </style>
    <style name="Theme.P1" parent="Base.Theme.P1">
        <item name="android:textColor">#ff0810</item>
        <item name="android:background">#aa9911</item>
    </style>
</resources>
```

MainActivity.java

```
package com.example.prac_1;

import android.os.Bundle;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;
public class MainActivity extends AppCompatActivity {
    @Override

    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        // Find the TextView by its ID and set its text
        TextView textView = findViewById(R.id.textView);
        textView.setText("My application");
```

```
    TextView textView2 = findViewById(R.id.textView2);
    textView2.setText(getString(R.string.t1));  }}
```

activity_main.xml

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:app="http://schemas.android.com/apk/res-auto"
   xmlns:tools="http://schemas.android.com/tools"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   tools:context=".MainActivity">
   <TextView
      android:id="@+id/textView"
      android:layout_width="@dimen/d1"
      android:layout_height="wrap_content"
      android:background="@color/d4"
      android:text="My application"
      android:textColor="@color/white"
      app:layout_constraintBottom_toBottomOf="parent"
      app:layout_constraintEnd_toEndOf="parent"
      app:layout_constraintStart_toStartOf="parent"
      app:layout_constraintTop_toTopOf="parent"
      app:layout_constraintVertical_bias="0.022" />

   <TextView
      android:id="@+id/textView2"
      android:layout_width="@dimen/d2"
      android:layout_height="wrap_content"
      android:background="@color/d1"
      android:rotationX="4"
      android:text="@string/t1"
      android:textColor="#212121"
      app:layout_constraintBottom_toBottomOf="parent"
      app:layout_constraintEnd_toEndOf="parent"
      app:layout_constraintStart_toStartOf="parent"
      app:layout_constraintTop_toTopOf="parent" />

   <ImageView
      android:id="@+id/imageView"
      android:layout_width="135dp"
      android:layout_height="180dp"
      app:srcCompat="@drawable/circle" />
```

```
    <ImageView
        android:id="@+id/imageView3"
        android:layout_width="136dp"
        android:layout_height="181dp"
        app:srcCompat="@drawable/random" />
</LinearLayout>
```

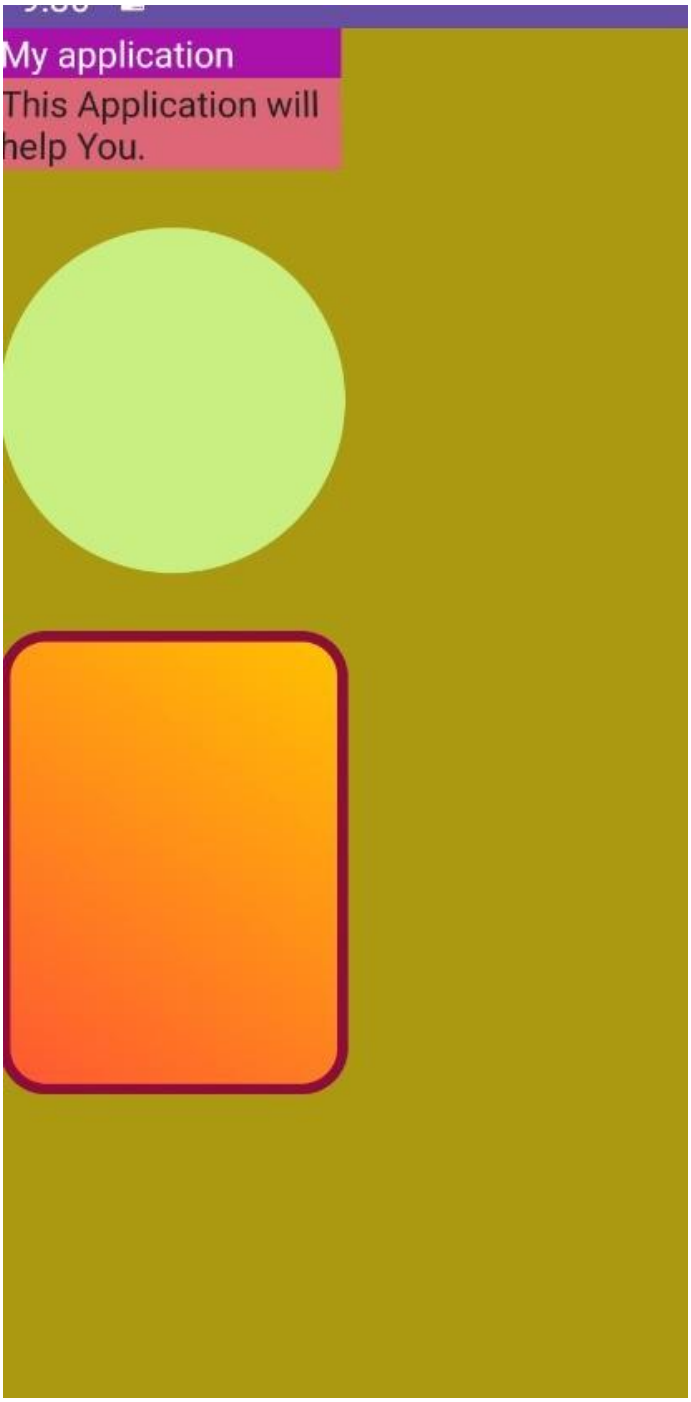
circle.xml (drawable file)

```
<?xml version="1.0" encoding="utf-8"?>
<selector xmlns:android="http://schemas.android.com/apk/res/android">
    <item>
        <shape android:shape="oval">
            <solid android:color="#c9ee82" />
            <size android:width="100dp" android:height="100dp" />
        </shape>
    </item>
</selector>
```

random.xml (drawable file)

```
<?xml version="1.0" encoding="utf-8"?>
<selector xmlns:android="http://schemas.android.com/apk/res/android">
    <item>
        <shape android:shape="rectangle">
            <corners android:radius="16dp" /> <!-- Rounded corners -->
            <gradient
                android:startColor="#FF5733"
                android:endColor="#FFC300"
                android:angle="45" /> <!-- Gradient color -->
            <stroke
                android:width="4dp"
                android:color="#900C3F" /> <!-- Border color and width -->
            <padding
                android:left="10dp"
                android:top="10dp"
                android:right="10dp"
                android:bottom="10dp" /> <!-- Padding -->
        </shape>

    </item>
</selector>
```

Output:



Practical 2

Programming Activities and fragments

Aim: Use Activity Life Cycle, Activity methods, Multiple Activities, Life Cycle of fragments and multiple fragments.

Files Used:

Project Name: Prac_2

Java File 1: MainActivity.java

Java File 2: MainActivity2.java

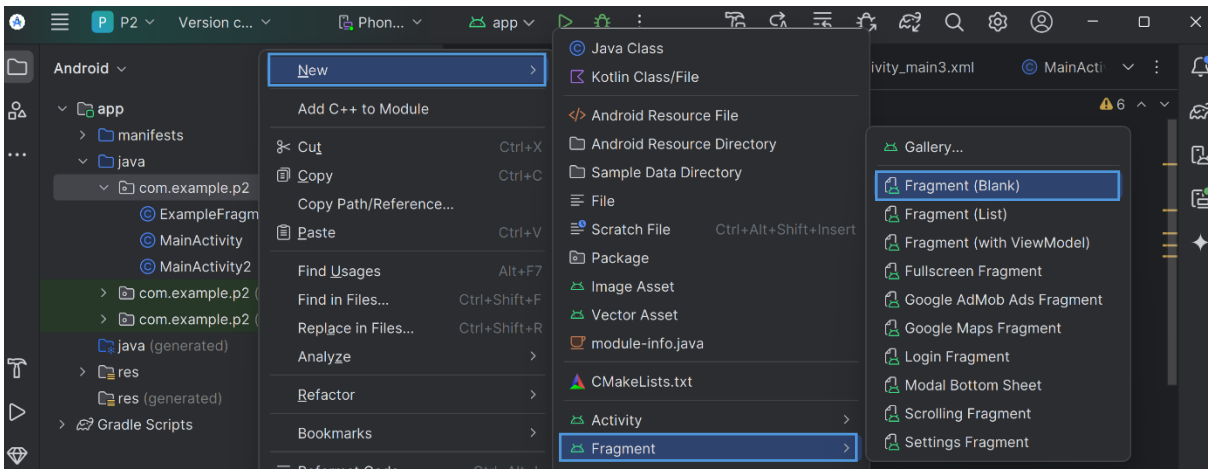
Fragment File: ExampleFragrment.java

XML File: activity_main.xml

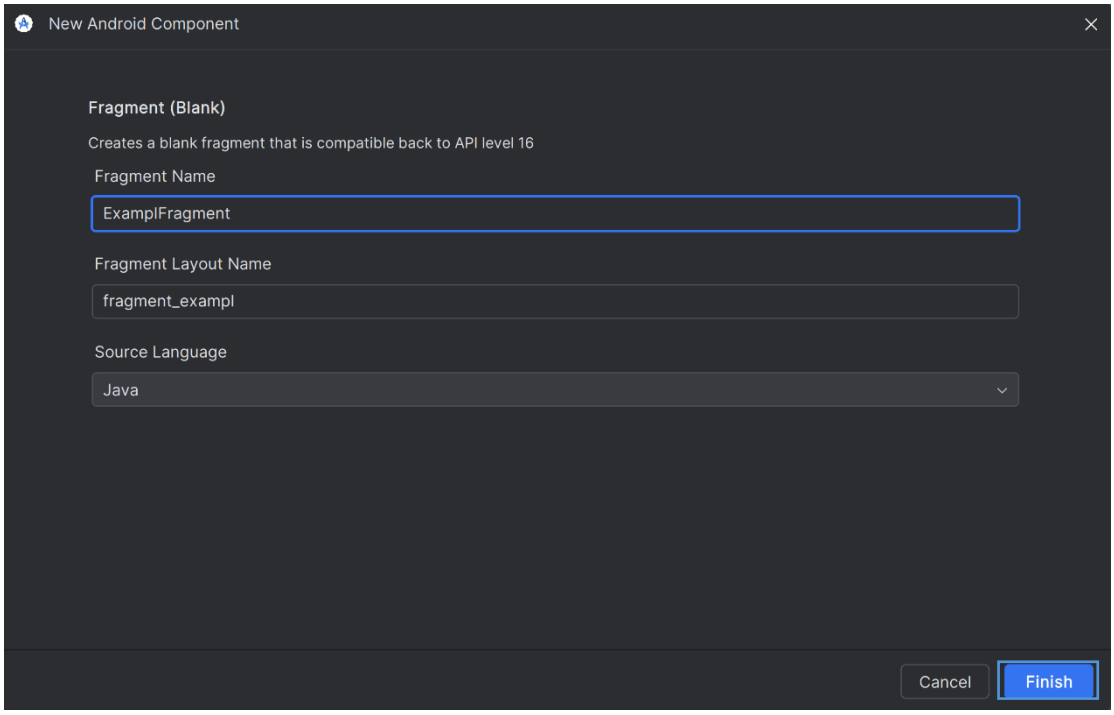
fragment_example.xml

Steps to make a Fragment File:

1. Right click on 'com.example.p2' folder → New → Select 'Fragment' → 'Fragment (Blank) File'.



2. Give name as 'ExampleFragment' → Finish.



Code: <u>MainActivity.java</u>

```
package com.example.prac_2;

import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;

import androidx.appcompat.app.AppCompatActivity;
import androidx.fragment.app.FragmentManager;
import androidx.fragment.app.FragmentTransaction;

public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
```

```
        setContentView(R.layout.activity_main);

        Button buttonNavigate = findViewById(R.id.button_navigate);
        buttonNavigate.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                Intent intent = new Intent(MainActivity.this, MainActivity2.class);
                startActivity(intent);
            }
        });

        if (savedInstanceState == null) {
            ExampleFragment fragment = new ExampleFragment();
            FragmentManager fragmentManager = getSupportFragmentManager();
            FragmentTransaction fragmentTransaction = fragmentManager.beginTransaction();
            fragmentTransaction.add(R.id.fragment_container, fragment);
            fragmentTransaction.commit();
        }
    }
}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">


    <Button
        android:id="@+id/button_navigate"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
```

```
        android:text="Go to Second Activity" />

    <FrameLayout
        android:id="@+id/fragment_container"
        android:layout_width="match_parent"
        android:layout_height="match_parent" />

</LinearLayout>
```

MainActivity2.java

```
package com.example.prac_2;

import android.os.Bundle;
import android.util.Log;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity2 extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main3);
        Log.d("SecondActivity", "onCreate invoked");
    }

    @Override
    protected void onStart() {
        super.onStart();
        Log.d("SecondActivity", "onStart invoked");
    }

    @Override
    protected void onResume() {
        super.onResume();
        Log.d("SecondActivity", "onResume invoked");
    }

    @Override
    protected void onPause() {
        super.onPause();
        Log.d("SecondActivity", "onPause invoked");
    }
```

```
    @Override
    protected void onStop() {
        super.onStop();
        Log.d("SecondActivity", "onStop invoked");
    }

    @Override
    protected void onDestroy() {
        super.onDestroy();
        Log.d("SecondActivity", "onDestroy invoked");
    }
}
```

ExampleFragment.java

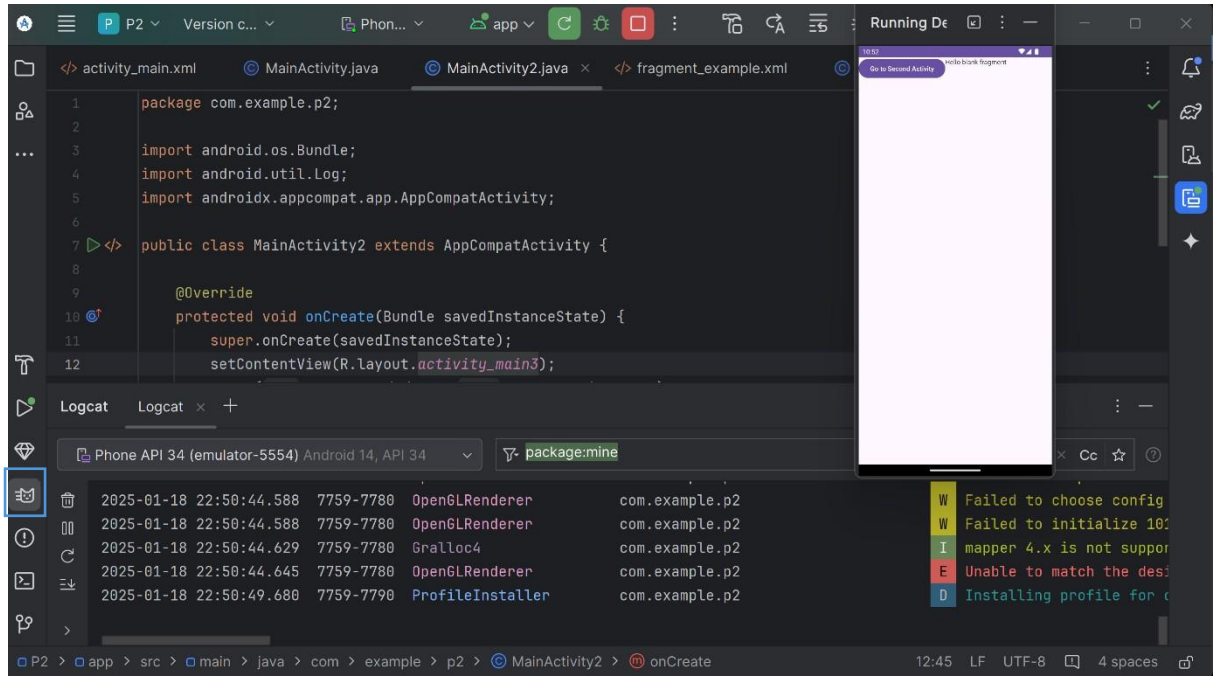
```
package com.example.prac_2;

import android.os.Bundle;
import android.util.Log;
import androidx.fragment.app.Fragment;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;

public class ExampleFragment extends Fragment {

    @Override
    public View onCreateView(LayoutInflater inflater, ViewGroup container,
                             Bundle savedInstanceState) {
        Log.d("ExampleFragment", "onCreateView invoked");
        return inflater.inflate(R.layout.fragment_example, container, false);
    }

    @Override
    public void onStart() {
        super.onStart();
        Log.d("ExampleFragment", "onStart invoked");
    }

    @Override
    public void onResume() {
        super.onResume();
        Log.d("ExampleFragment", "onResume invoked");
    }
```

```
    @Override
    public void onPause() {
        super.onPause();
        Log.d("ExampleFragment", "onPause invoked");
    }

    @Override
    public void onStop() {
        super.onStop();
        Log.d("ExampleFragment", "onStop invoked");
    }

    @Override
    public void onDestroyView() {
        super.onDestroyView();
        Log.d("ExampleFragment", "onDestroyView invoked");
    }
}
```

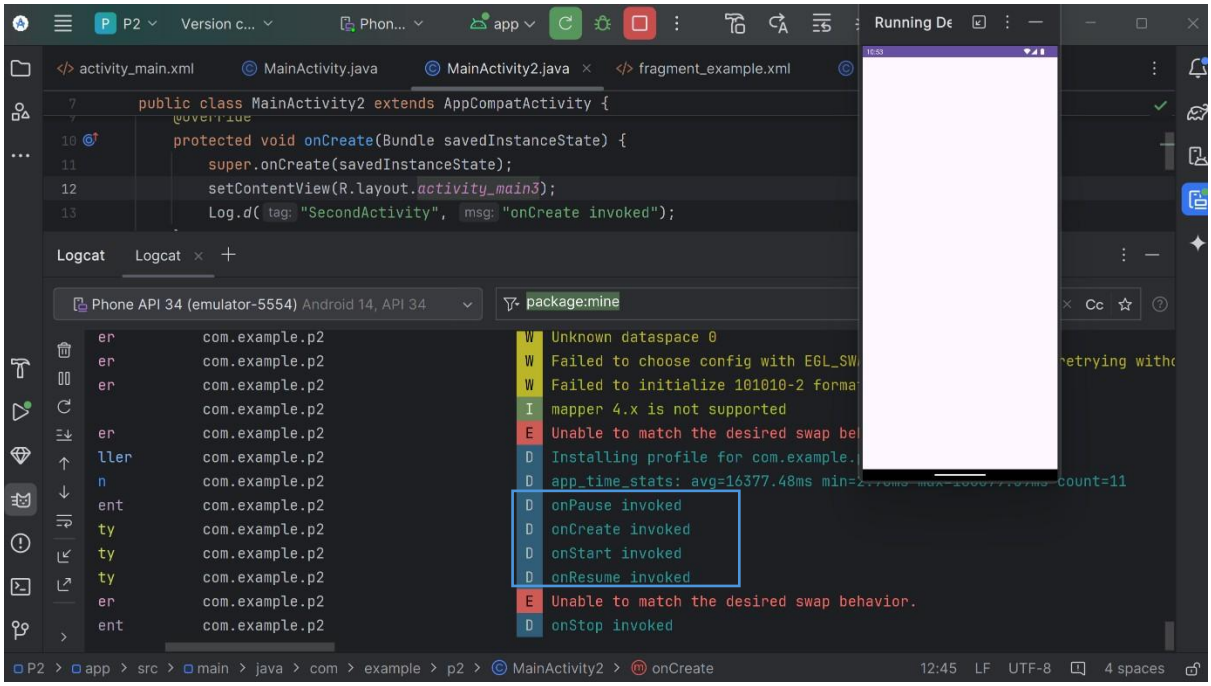
fragment_example.xml

```
<?xml version="1.0" encoding="utf-8"?>
<FrameLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".ExampleFragment">

    <!-- TODO: Update blank fragment layout -->
    <TextView
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:text="@string/hello_blank_fragment" />

</FrameLayout>
```

Output:



When clicked on 'Go to Second Activity' button



Note: onCreate() is called when the activity is first created. **onStart()** is called when the activity becomes visible to the user. **onResume()** is called when the activity starts interacting with the user. **onPause()** is called when the activity is partially obscured by another. **onStop()** is called when the activity is no longer visible to the user. **onDestroy()** is called before the activity is destroyed.

Practical 3

Programs related to different Layouts

Aim: Use Coordinate, Linear, Relative, Table, Absolute, Frame, List View, Grid View.

1. **Linear Layout:**

Files Used:

Project Name: Prac_3

Java File: MainActivity.java

XML File: activity_main.xml

Code: MainActivity.java

```
package com.example.prac_3;

import android.os.Bundle;
import android.widget.Button;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        Button button1 = findViewById(R.id.button1);
        Button button2 = findViewById(R.id.button2);
        button1.setOnClickListener(v -> Toast.makeText(MainActivity.this, "Button 1
clicked", Toast.LENGTH_SHORT).show());
        button2.setOnClickListener(v -> Toast.makeText(MainActivity.this, "Button 2
clicked", Toast.LENGTH_SHORT).show());
    }}
```

activity_main.xml

```
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical">
    <Button
        android:id="@+id/button1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Button 1" />
    <Button
        android:id="@+id/button2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Button 2" />
</LinearLayout>
```

Output:

2. **Relative Layout:**

Code: MainActivity2.java

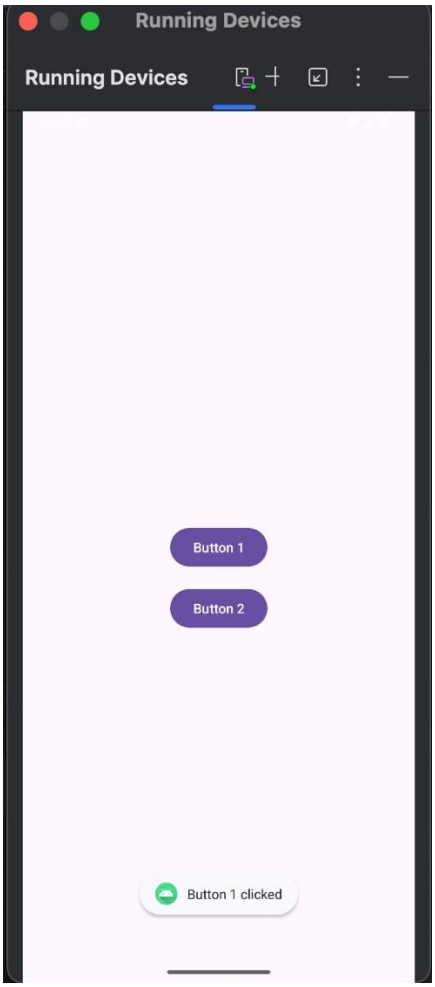
```
package com.example.prac_3;
import android.os.Bundle;
import android.widget.Button;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        Button button1 = findViewById(R.id.button1);
        Button button2 = findViewById(R.id.button2);

        button1.setOnClickListener(v -> Toast.makeText(MainActivity.this, "Button 1
clicked", Toast.LENGTH_SHORT).show());
        button2.setOnClickListener(v -> Toast.makeText(MainActivity.this, "Button 2
clicked", Toast.LENGTH_SHORT).show());
    }
}
```

activity_main2.xml

```
<RelativeLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="16dp">
```

```
    <Button
        android:id="@+id/button1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Button 1"
        android:layout_centerInParent="true" />
    <Button
        android:id="@+id/button2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Button 2"
        android:layout_below="@id/button1"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="16dp" />
</RelativeLayout>
```

Output:



3. **Table Layout:**

Code: MainActivity3.java

```
package com.example.prac_3;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    private EditText usernameEditText;
    private EditText passwordEditText;
    private Button loginButton;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        // Initialize the views
        usernameEditText = findViewById(R.id.username);
        passwordEditText = findViewById(R.id.password);
        loginButton = findViewById(R.id.login_button);

        // Set the login button click listener
        loginButton.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                String username = usernameEditText.getText().toString();
                String password = passwordEditText.getText().toString();
```

```
            // Check for empty fields
            if (username.isEmpty() || password.isEmpty()) {
                Toast.makeText(MainActivity.this, "Please fill in both fields",
Toast.LENGTH_SHORT).show();
            } else {
                // Here, you can check username and password (for example, static check)
                if (username.equals("tyitb") && password.equals("1234")) {
                    Toast.makeText(MainActivity.this, "Login Successful",
Toast.LENGTH_SHORT).show();
                } else {
                    Toast.makeText(MainActivity.this, "Invalid credentials",
Toast.LENGTH_SHORT).show();
                }
            }
        }
    });
  }
}
```

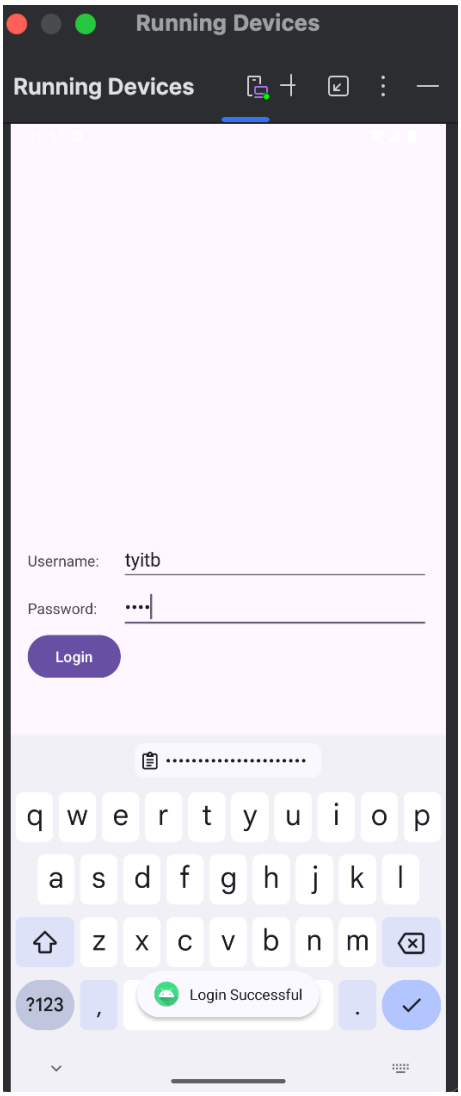
activity_main3.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="16dp">

    <TableLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
```

```
    android:stretchColumns="1">

    <TableRow>
        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Username:"
            android:paddingEnd="8dp"
            android:labelFor="@id/username"/>

        <EditText
            android:id="@+id/username"
            android:layout_width="0dp"
            android:layout_height="wrap_content"
            android:layout_weight="1"
            android:hint="Enter your username"
            android:inputType="text" />
    </TableRow>

    <TableRow>
        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Password:"
            android:paddingEnd="8dp"
            android:labelFor="@id/password"/>

        <EditText
            android:id="@+id/password"
            android:layout_width="0dp"
            android:layout_height="wrap_content"
            android:layout_weight="1"
            android:hint="Enter your password"
            android:inputType="textPassword" />
```

```
        </TableRow>

        <TableRow>
            <Button
                android:id="@+id/login_button"
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
                android:text="Login" />
        </TableRow>
    </TableLayout>
</LinearLayout>
```

Output:



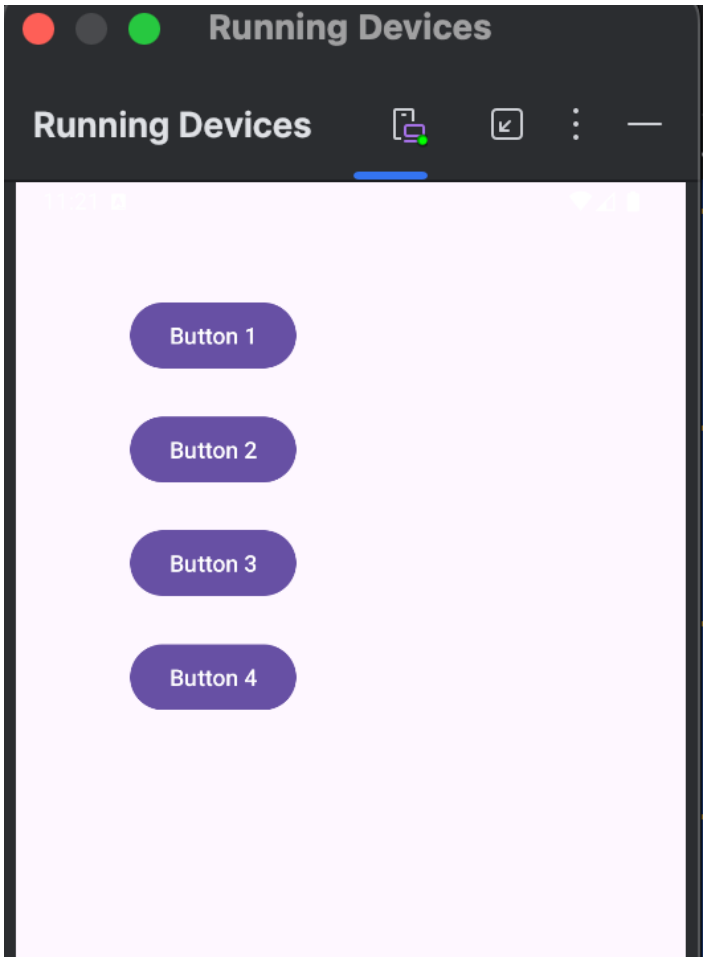
4. Absolute Layout:

Code: activity_main4.xml

```
<?xml version="1.0" encoding="utf-8"?>
<AbsoluteLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="20dp">
    <Button
        android:id="@+id/button1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Button 1"
        android:layout_x="50dp"
        android:layout_y="50dp" />
    <Button
        android:id="@+id/button2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Button 2"
        android:layout_x="50dp"
        android:layout_y="120dp" />
    <Button
        android:id="@+id/button3"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Button 3"
        android:layout_x="50dp"
        android:layout_y="190dp" />
    <Button
        android:id="@+id/button4"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
```

```
        android:text="Button 4"
        android:layout_x="50dp"
        android:layout_y="260dp" />
</AbsoluteLayout>
```

Output:



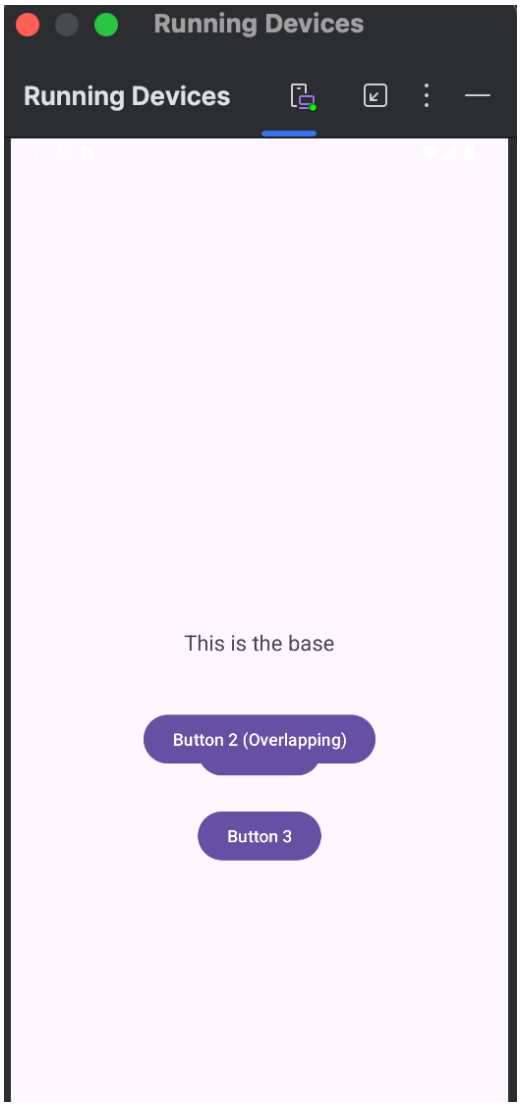
5. Frame Layout:

Code: activity_main5.xml

```
<FrameLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <TextView
        android:id="@+id/baseText"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="This is the base"
        android:layout_gravity="center"
        android:textSize="18sp"
        android:paddingBottom="80dp" />

    <Button
        android:id="@+id/button1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Button 1"
        android:layout_gravity="center"
        android:layout_marginTop="50dp" />

    <Button
        android:id="@+id/button2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Button 2 (Overlapping)"
        android:layout_gravity="center"
        android:layout_marginTop="40dp" />
```

```
    <Button
        android:id="@+id/button3"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Button 3"
        android:layout_gravity="center"
        android:layout_marginTop="120dp" />

</FrameLayout>
```

Output:



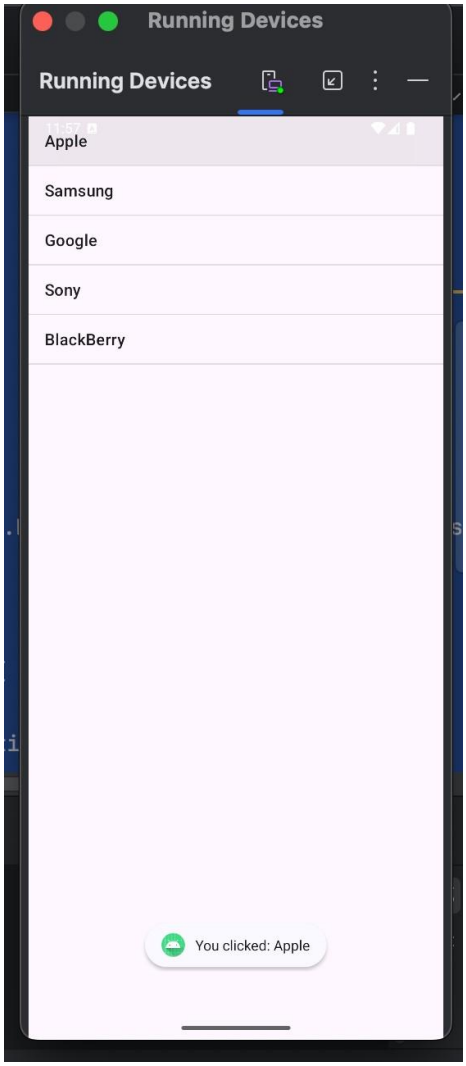
6. List View Layout:

Code: MainActivity6.java

```
package com.example.prac_3;
import android.os.Bundle;
import android.view.View;
import android.widget.AdapterView;
import android.widget.ArrayAdapter;
import android.widget.ListView;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    private ListView listView;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        // Initialize ListView
        listView = findViewById(R.id.listView);
        // Create an array of labels
        String[] labels = {"Apple", "Samsung", "Google", "Sony", "BlackBerry"};

        // Create an ArrayAdapter to bind the array to the ListView
        ArrayAdapter<String> adapter = new ArrayAdapter<>(this,
android.R.layout.simple_list_item_1, labels);
        listView.setAdapter(adapter);

        // Set an OnItemClickListener for the ListView
        listView.setOnItemClickListener(new AdapterView.OnItemClickListener() {
            @Override
            public void onItemClick(AdapterView<?> parent, View view, int position, long
id) {
```

```
            // Get the label text from the clicked item
            String labelName = (String) parent.getItemAtPosition(position);

            // Show a Toast with the name of the clicked label
            Toast.makeText(MainActivity.this, "You clicked: " + labelName,
Toast.LENGTH_SHORT).show();
        }
    });
  }
}
```

Output:



7. Grid View Layout:

Code: MainActivity7.java

```
package com.example.prac_3;
import android.os.Bundle;
import android.view.View;
import android.view.ViewGroup;
import android.widget.GridView;
import android.widget.ImageView;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    private GridView gridView;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        gridView = findViewById(R.id.gridView);
        int[] imageIds = {
                R.mipmap.ic_launcher,
                R.mipmap.ic_launcher,
                R.mipmap.ic_launcher,
                R.mipmap.ic_launcher,
                R.mipmap.ic_launcher,
                R.mipmap.ic_launcher
        };
        gridView.setAdapter(new ImageAdapter(imageIds));
        gridView.setOnItemClickListener((parent, view, position, id) -> {
            Toast.makeText(MainActivity.this, "You clicked: Image " + (position + 1),
Toast.LENGTH_SHORT).show();
        });
    }
```

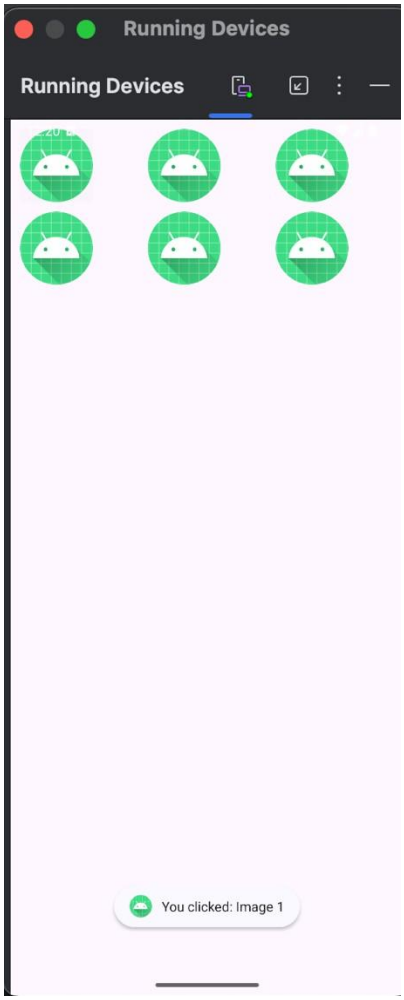
```
public class ImageAdapter extends android.widget.BaseAdapter {
    private int[] imageIds;
    public ImageAdapter(int[] imageIds) {
        this.imageIds = imageIds;
    }
    @Override
    public int getCount() {
        return imageIds.length;
    }
    @Override
    public Object getItem(int position) {
        return imageIds[position];
    } @Override
    public long getItemId(int position) {
        return position;
    } @Override
    public View getView(int position, View convertView, ViewGroup parent) {
        ImageView imageView = new ImageView(MainActivity.this);
        imageView.setImageResource(imageIds[position]);  // Set the image resource for
each item
        imageView.setLayoutParams(new ViewGroup.LayoutParams(200, 200)); // Set
size for images
        imageView.setScaleType(ImageView.ScaleType.CENTER_CROP); // Scale
image properly
        return imageView;
    }}}
```

activity_main7.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
```

```
    android:layout_height="match_parent"
    android:orientation="vertical"
    tools:ignore="ExtraText">
    <GridView
        android:id="@+id/gridView"
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:numColumns="3"
    android:verticalSpacing="10dp"
    android:horizontalSpacing="10dp"
    android:padding="10dp"
    android:stretchMode="columnWidth"
    android:columnWidth="90dp" />
</LinearLayout>
```

Output:



Practical 4

Programming UI elements

Part A: -

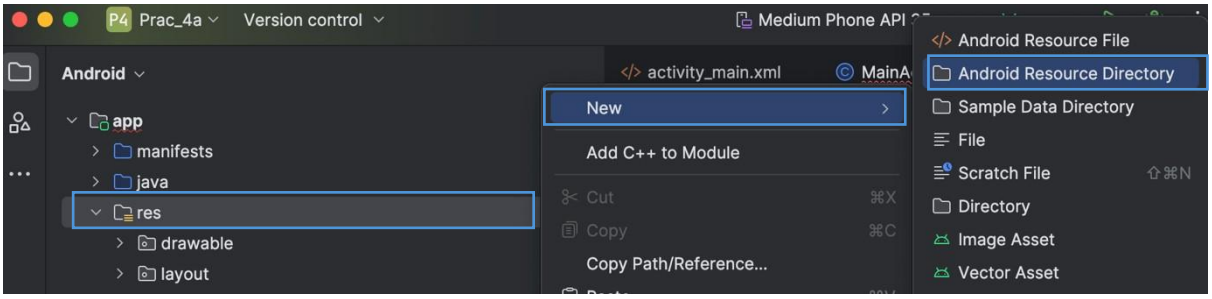
Aim: Use AppBar

Files Used:

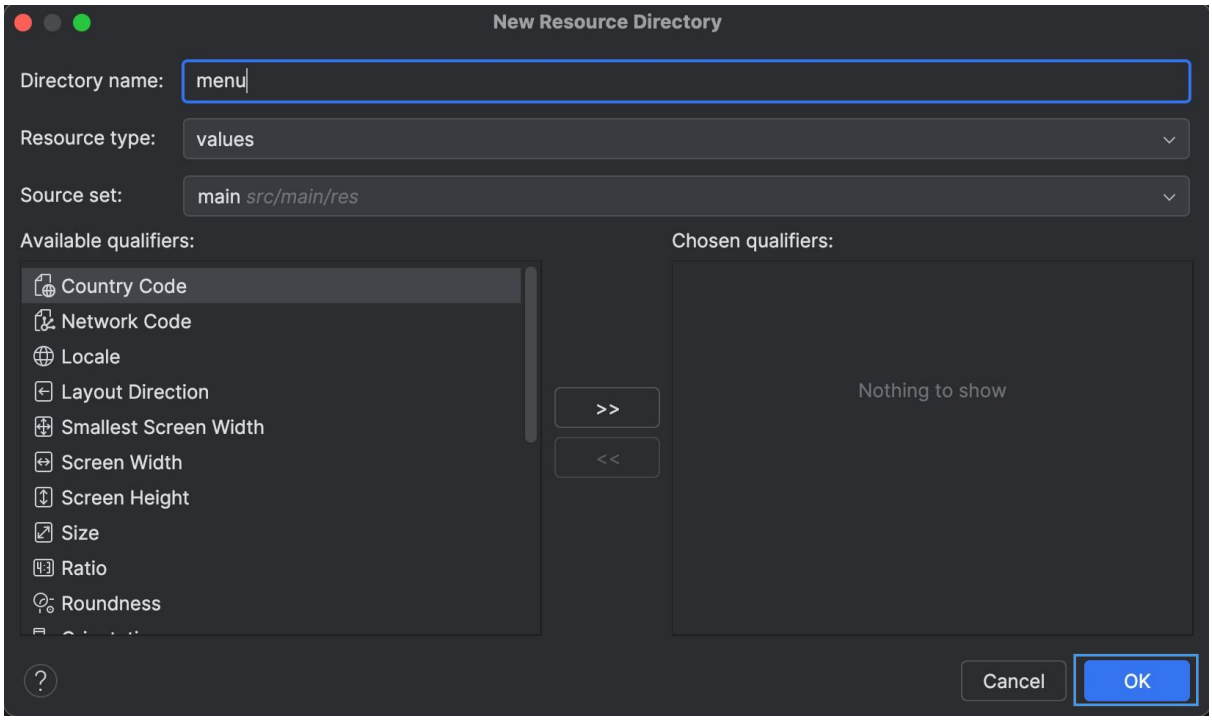
Project Name: Prac_4a
Java File: MainActivity.java
XML File: activity_main.xml
Android Resource File: menu.xml

Steps to make Android Resource Directory:

1. Right click on 'res' file → 'New' → select 'Android Resource Directory'.

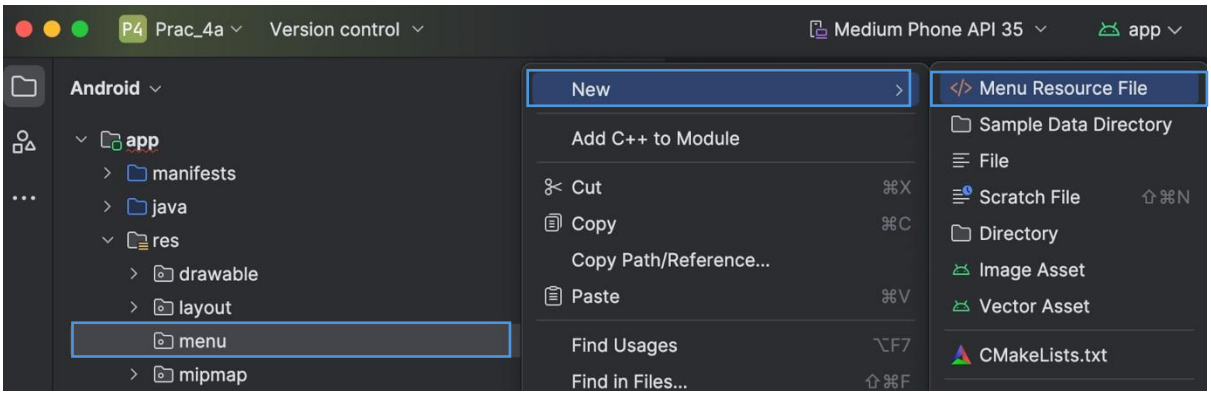


2. Name it as ‘menu’ → Ok.

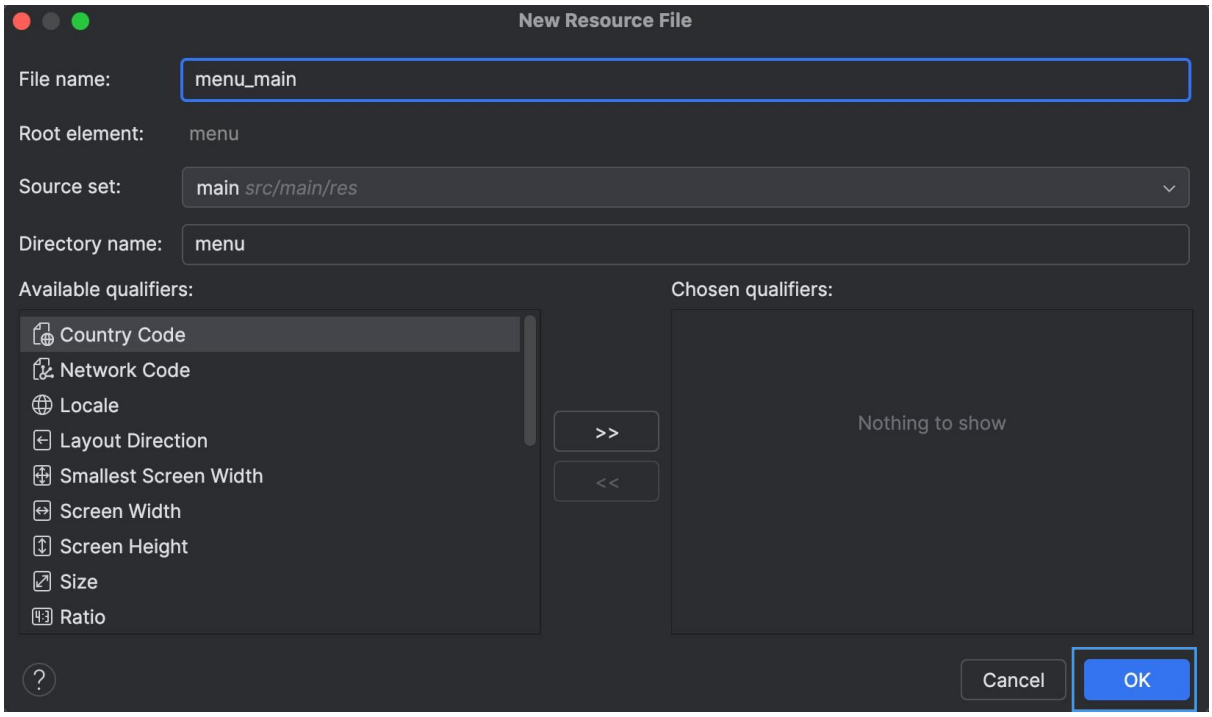


Steps to make Android Resource File:

1. Right click on ‘res’ file → ‘New’ → select ‘Android Resource Directory’



2. Name it as ‘menu_main’ → Ok.



Code: <u>menu_main.xml</u>

```
<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto">

    <item
        android:id="@+id/action_settings1"
        android:title="Settings"
        android:orderInCategory="100" />

    <item
        android:id="@+id/action_settings2"
        android:title="Settings1"
        android:orderInCategory="100" />
</menu>
```

activity_main.xml

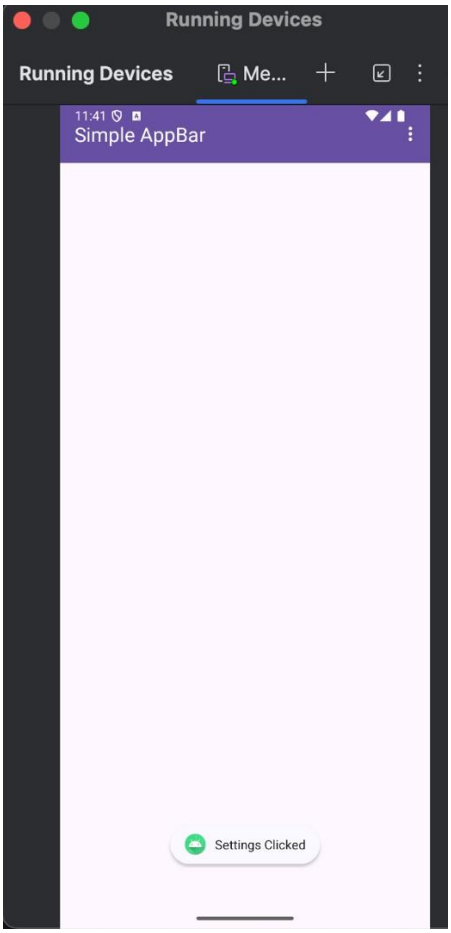
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    tools:context=".MainActivity">

      <androidx.appcompat.widget.Toolbar
        android:id="@+id/toolbar"
        android:layout_width="match_parent"
        android:layout_height="?attr/actionBarSize"
        android:background="?attr/colorPrimary"
        android:elevation="4dp"
        android:theme="@style/ThemeOverlay.AppCompat.Dark.ActionBar"
        app:title="Simple AppBar"/>

</LinearLayout>
```

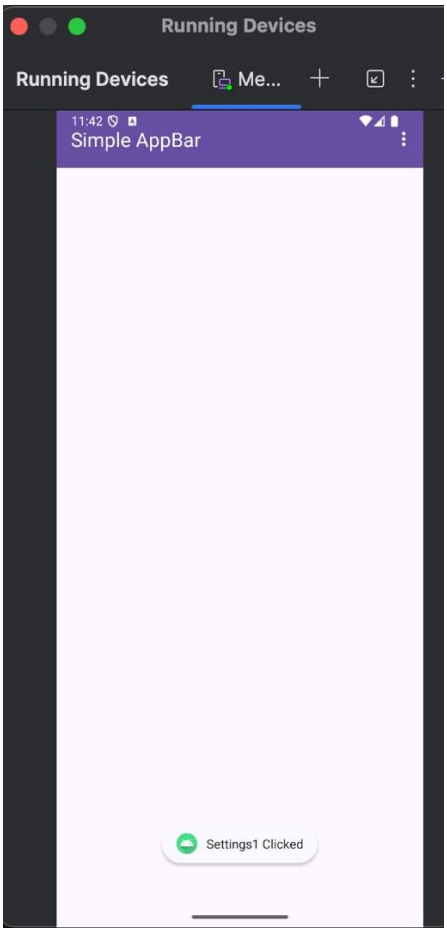
ActivityMain.java

```
package com.example.prac_4a;

import androidx.annotation.NonNull;
import androidx.appcompat.app.AppCompatActivity;
import androidx.appcompat.widget.Toolbar;
import android.os.Bundle;
import android.view.Menu;
import android.view.MenuItem;
```

```
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        // Initialize Toolbar
        Toolbar toolbar = findViewById(R.id.toolbar);
        setSupportActionBar(toolbar);
    }

    @Override
    public boolean onCreateOptionsMenu(Menu menu) {
        getMenuInflater().inflate(R.menu.menu_main, menu);
        return true;
    }

    @Override
    public boolean onOptionsItemSelected(@NonNull MenuItem item) {
        int id = item.getItemId();
        if (id == R.id.action_settings1) {
            Toast.makeText(this, "Settings Clicked", Toast.LENGTH_SHORT).show();
            return true;
        } else if (id == R.id.action_settings2) {
            Toast.makeText(this, "Settings1 Clicked", Toast.LENGTH_SHORT).show();
            return true;
        }
        return super.onOptionsItemSelected(item);
    }
}
```

Output:


Running Devices
Running Devices
Me...
11:41
Simple AppBar
Settings Clicked


Running Devices
Running Devices
Me...
11:42
Simple AppBar
Settings1 Clicked

Part B: -

Aim: Use Fragments

Files Used:

Project Name: Prac_4b

Java File: MainActivity.java

XML File: activity_main.xml

Fragment File: BlankFragment.java & fragment_blank.xml

Steps to make a Fragment File same as 'Practical 2'

Code: fragment_blank.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">

    <Button
        android:id="@+id/btnDisplayMessage"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Click Me!" />

    <TextView
        android:id="@+id/textViewMessage"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Message will appear here"
        android:paddingTop="20dp"
        android:textSize="18sp" />

</LinearLayout>
```

BlankFragment.java

```
package com.example.prac_4b;

import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.Button;
import android.widget.TextView;
import androidx.fragment.app.Fragment;

public class BlankFragment extends Fragment {

    @Override
    public View onCreateView(LayoutInflater inflater, ViewGroup container,
                             Bundle savedInstanceState) {
        View rootView = inflater.inflate(R.layout.fragment_blank, container, false);

        Button btnDisplayMessage = rootView.findViewById(R.id.btnDisplayMessage);
        TextView textViewMessage = rootView.findViewById(R.id.textViewMessage);

        btnDisplayMessage.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                textViewMessage.setText("Hello from Fragment");
            }
        });

        return rootView;
    }
}
```

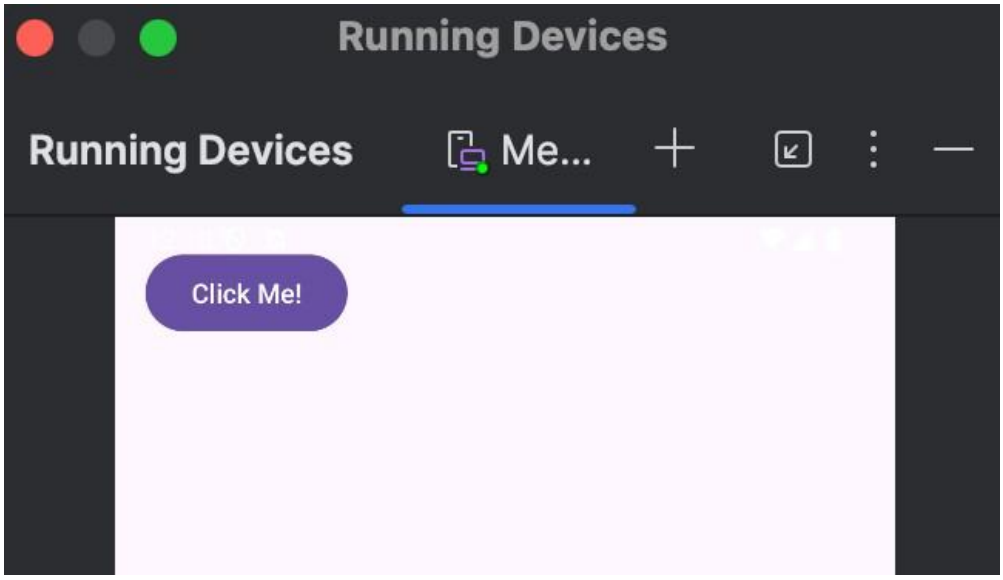
activity_main.xml

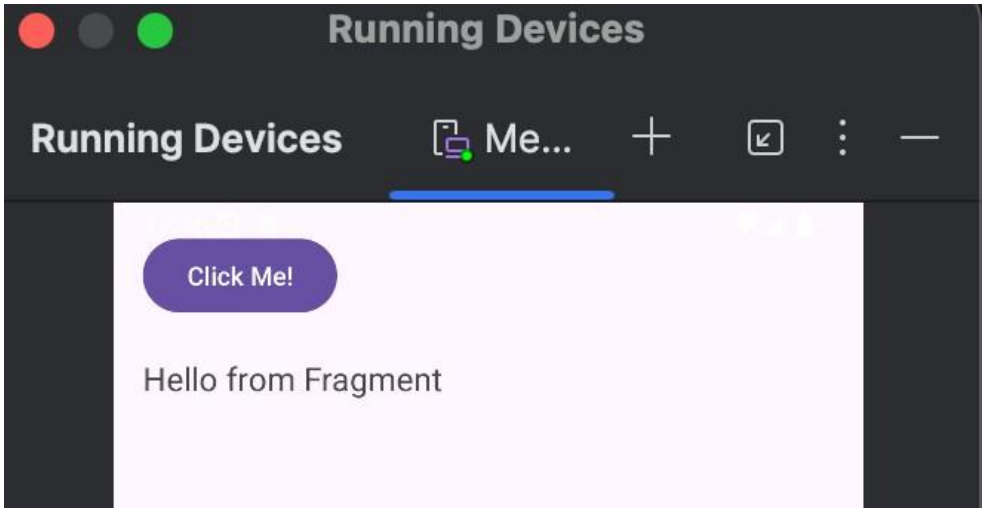
```
<?xml version="1.0" encoding="utf-8"?>
<FrameLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">
</FrameLayout>
```

MainActivity.java

```
package com.example.prac_4b;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;
import androidx.fragment.app.FragmentTransaction;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        // Add the fragment dynamically to the activity
        if (savedInstanceState == null) {
            BlankFragment sampleFragment = new BlankFragment();
            FragmentTransaction transaction =
getSupportFragmentManager().beginTransaction();
            transaction.replace(android.R.id.content, sampleFragment);  // Replace with
FrameLayout container
            transaction.commit();
        }
    }
}
```

Output:


Running Devices
Running Devices
Me...
Click Me!


Running Devices
Running Devices
Me...
Click Me!
Hello from Fragment

Part C: -

Aim: Use UI components

Files Used:

Project Name: Prac_4c

Java File: MainActivity.java

XML File: activity_main.xml

Code: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    tools:context=".MainActivity">

    <Button
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:id="@+id/btnSave"
        android:text="Save"
        android:layout_alignParentTop="true"/>
    <ImageButton
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:id="@+id/ib"
        android:src="@mipmap/ic_launcher"/>
    <EditText
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:hint="edit text"
        android:id="@+id/et"
        />
    <CheckBox
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:id="@+id/cb1"
```

```
        android:text="Auto Save"/>

    <CheckBox
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:id="@+id/cbst"
        android:text=""
        style="?android:attr/starStyle"/>


    <RadioGroup
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:id="@+id/rg">
        <RadioButton
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:id="@+id/rb1"
            android:text="Option1"/>
        <RadioButton
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:id="@+id/rb2"
            android:text="Option2"/>
    </RadioGroup>
    <ToggleButton
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:id="@+id/tb"/>
</LinearLayout>
```

ActivityMain.java

```
package com.example.prac_4c;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.Display;
import android.view.View;
```

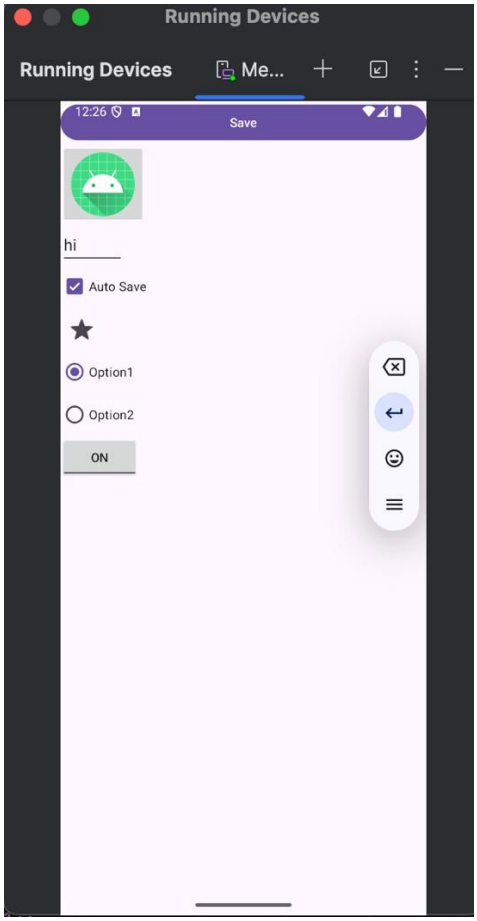
```
import android.widget.Button;
import android.widget.CheckBox;
import android.widget.CompoundButton;
import android.widget.EditText;
import android.widget.ImageButton;
import android.widget.RadioButton;
import android.widget.ToggleButton;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        Button btsave=(Button) findViewById(R.id.btnSave);
        ImageButton imgb=(ImageButton) findViewById(R.id.ib);
        EditText etext=(EditText) findViewById(R.id.et);
        CheckBox cb1=(CheckBox) findViewById(R.id.cb1);
        CheckBox cbst=(CheckBox) findViewById(R.id.cbst);
        RadioButton rb1=(RadioButton) findViewById(R.id.rb1);
        RadioButton rb2=(RadioButton) findViewById(R.id.rb2);
        ToggleButton tbutton =(ToggleButton) findViewById(R.id.tb);
        btsave.setOnClickListener(new View.OnClickListener()
        {
            @Override
            public void onClick(View v)
            {
                DisplayToast("Save Button clicked");
            }
        });

        imgb.setOnClickListener(new View.OnClickListener() {
            @Override
```

```
    public void onClick(View v) {
        DisplayToast("ImageButton Clicked");
    }
});

cb1.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        if(((CheckBox)v).isChecked())
            DisplayToast("CheckBox Option1 is checked");
        else
            DisplayToast(" CheckBox  unchecked");
    }
});

cbst.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        if(((CheckBox)v).isChecked())
            DisplayToast("CheckBox Option2 is checked");
        else
            DisplayToast(" CheckBox  unchecked");
    }
});
cbst.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        if(((CheckBox)v).isChecked())
            DisplayToast("CheckBox Option2 is checked");
        else
            DisplayToast(" CheckBox  unchecked");
    }
});
rb1.setOnCheckedChangeListener(new CompoundButton.OnCheckedChangeListener()
```

```
{
    @Override
    public void onCheckedChanged(CompoundButton buttonView,boolean ischecked)
    {
        if(rb1.isChecked()) {
            DisplayToast("RadioButton 1 is checked");
        }
        else
            DisplayToast("RadioButton 1 is unchecked");

    }
});

rb2.setOnCheckedChangeListener(new CompoundButton.OnCheckedChangeListener()
{
    @Override
    public void onCheckedChanged(CompoundButton buttonView,boolean ischecked)
    {
        if(rb2.isChecked()) {
            DisplayToast("RadioButton 2 is checked");
        }
        else
            DisplayToast("RadioButton 2 is unchecked");

    }
});
tbutton.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        if(tbutton.isChecked()) {
            DisplayToast("ToggleButton is on");
        }
        else
            DisplayToast("ToggleButton is off");
```

```
            }
        });
    }
    public void DisplayToast(String s)
    {
        Toast.makeText(this,s,Toast.LENGTH_SHORT).show();
    }
}
```

Output:



Practical 5

Perform menus, dialog, dialog fragments.

Part A: -

Aim: Use Menus

Files Used:

Project Name: Prac_5a

Java File: MainActivity.java

XML File: themes.xml

Android Resource File: my_menu.xml

Steps to make Android Resource Directory & File are same as 'Practical 4a'

Code: themes.xml

```
<resources xmlns:tools="http://schemas.android.com/tools">
   <!-- Base application theme. -->
   <style name="Base.Theme.Menu" parent="Theme.AppCompat.Light.DarkActionBar">
      <!-- Customize your light theme here. -->
      <!-- <item name="colorPrimary">@color/my_light_primary</item> -->
   </style>
   <style name="Theme.Menu" parent="Base.Theme.Menu" />
</resources>
```

my_menu.xml

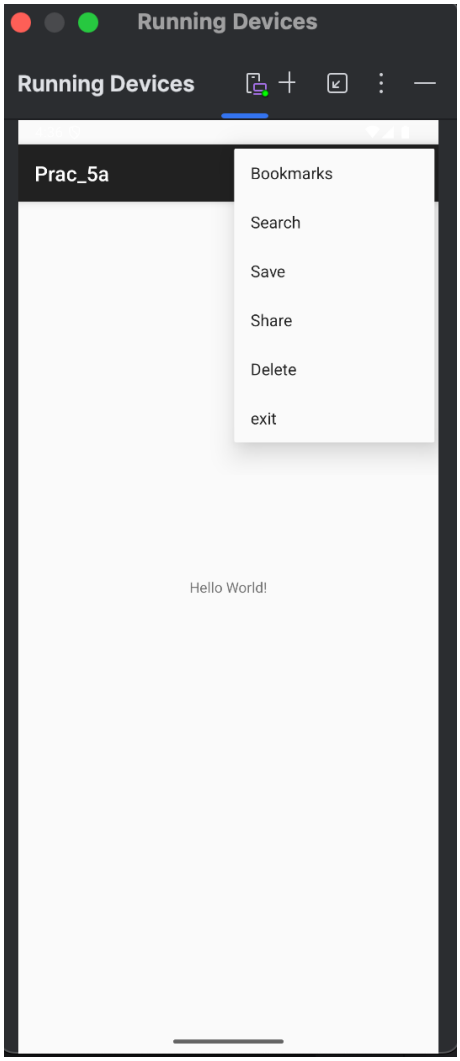
```
<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android">
    <item
        android:id="@+id/newb"
        android:title="Bookmarks"/>
    <item
        android:id="@+id/search"
        android:title="Search"/>
    <item
        android:id="@+id/save"
        android:title="Save"/>
    <item
        android:id="@+id/Share"
        android:title="Share"/>
    <item
        android:id="@+id/delete"
        android:title="Delete"/>
    <item
        android:id="@+id/exit"
        android:title="exit"/>
</menu>
```

MainActivity.java

```
package com.example.prac_5a;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.Menu;
import android.view.MenuInflater;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }

    @Override
    public boolean onCreateOptionsMenu(Menu menu) {
```

```
        // Inflating the menu resource file
        MenuInflater mi = getMenuInflater();
        mi.inflate(R.menu.my_menu, menu);
        return true;
    }
}
```

Output:



Part B: -

Aim: Use Dialog

Files Used:

Project Name: Prac_5b
Java File: MainActivity.java
XML File: activity_main.xml

Code: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    tools:context=".MainActivity">

    <Button
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:id="@+id/button_dialog"
        android:onClick="onClickDialog"
        android:text="Click to display an AlertDialog"/>

    <Button
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:id="@+id/button_progressdialog"
```

```
        android:onClick="onClickProgressDialog"
        android:text="Click to display a ProgressDialog"/>

    <Button
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:id="@+id/button_datedialog"
        android:onClick="onClickDateDialog"
        android:text="Click to display a DatePickerDialog"/>

    <Button
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:id="@+id/button_timedialog"
        android:onClick="onClickTimeDialog"
        android:text="Click to display a TimePickerDialog"/>

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:id="@+id/textView_date"
        android:textSize="50dp"/>

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:id="@+id/textView_time"
        android:textSize="50dp"/>

</LinearLayout>
```

MainActivity.java

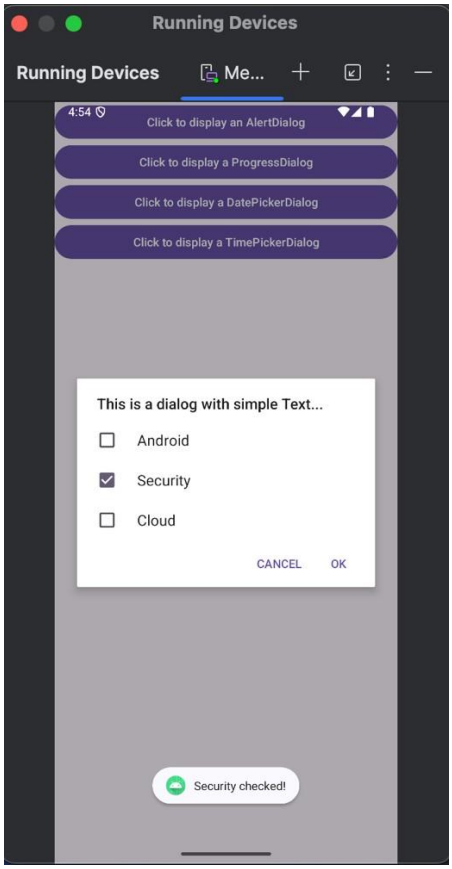
```
package com.example.prac_5b;

import android.app.DatePickerDialog;
import android.app.Dialog;
import android.app.ProgressDialog;
import android.app.TimePickerDialog;
import android.content.DialogInterface;
import android.os.Bundle;
import android.view.View;
import android.widget.DatePicker;
import android.widget.TextView;
import android.widget.TimePicker;
import android.widget.Toast;
import androidx.appcompat.app.AlertDialog;
import androidx.appcompat.app.AppCompatActivity;
import java.util.Calendar;

public class MainActivity extends AppCompatActivity {
    CharSequence[] items = {"Android", "Security", "Cloud"};
    boolean[] itemsChecked = new boolean[items.length];
    TextView DateDisplay, TimeDisplay;
    int mYear, mMonth, mDay, mHour, mMinute;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        DateDisplay = findViewById(R.id.textView_date);
        TimeDisplay = findViewById(R.id.textView_time);
    }
    public void onClickDialog(View v) {
        AlertDialog.Builder builder = new AlertDialog.Builder(this);
```

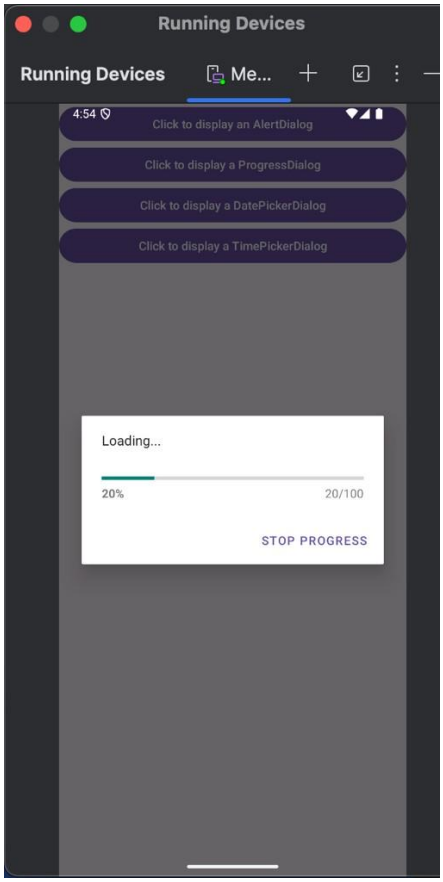
```
        builder.setTitle("This is a dialog with simple Text...");
        builder.setMultiChoiceItems(items, itemsChecked, new
DialogInterface.OnMultiChoiceClickListener() {
            public void onClick(DialogInterface dialog, int id, boolean isChecked) {
                Toast.makeText(getBaseContext(), items[id] + (isChecked ? " checked!" : "
unchecked!"), Toast.LENGTH_SHORT).show();
            }
        });
        builder.setPositiveButton("OK", new DialogInterface.OnClickListener() {
            @Override
            public void onClick(DialogInterface dialog, int id) {
                Toast.makeText(getBaseContext(), "OK Clicked",
Toast.LENGTH_SHORT).show();
            }
        });
        builder.setNegativeButton("CANCEL", new DialogInterface.OnClickListener() {
            @Override
            public void onClick(DialogInterface dialog, int id) {
                Toast.makeText(getBaseContext(), "CANCEL Clicked",
Toast.LENGTH_SHORT).show();
            }
        });
        AlertDialog dialog = builder.create();
        dialog.show();
    }
    @SuppressWarnings("deprecation")
    public void onClickProgressDialog(View v) {
        final ProgressDialog pDialog = new ProgressDialog(this);
        pDialog.setProgressStyle(ProgressDialog.STYLE_HORIZONTAL);
        pDialog.setMessage("Loading...");
        pDialog.incrementProgressBy(20);
        pDialog.setButton(Dialog.BUTTON_POSITIVE, "Stop Progress", new
DialogInterface.OnClickListener() {
            public void onClick(DialogInterface dialog, int id) {
```

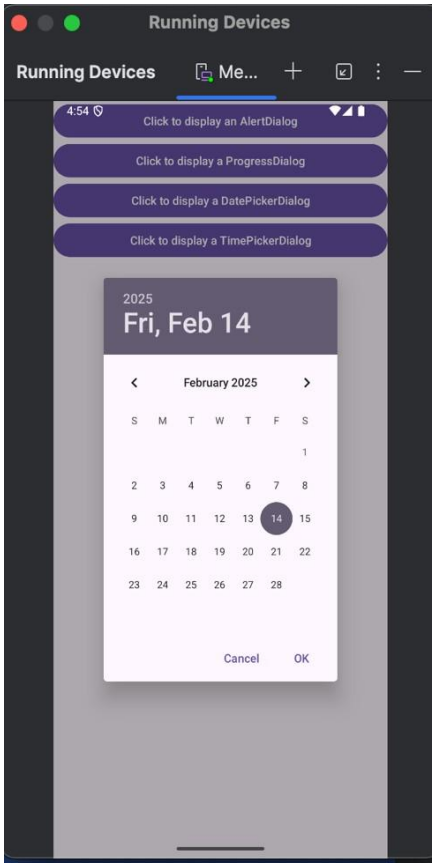
```
            pDialog.dismiss();
        }
    });

    pDialog.show();
  }
  public void onClickDateDialog(View v) {
    Calendar c = Calendar.getInstance();
    mYear = c.get(Calendar.YEAR);
    mMonth = c.get(Calendar.MONTH);
    mDay = c.get(Calendar.DAY_OF_MONTH);
    DatePickerDialog.OnDateSetListener mDateSetListener = new
DatePickerDialog.OnDateSetListener() {
        public void onDateSet(DatePicker view, int year, int month, int day) {
            DateDisplay.setText("Date: " + (month + 1) + "-" + day + "-" + year);
        }
    };
    DatePickerDialog dateDialog = new DatePickerDialog(this, mDateSetListener, mYear,
mMonth, mDay);
    dateDialog.show();
  }
  public void onClickTimeDialog(View v) {
    Calendar c = Calendar.getInstance();
    mHour = c.get(Calendar.HOUR_OF_DAY);
    mMinute = c.get(Calendar.MINUTE);
    TimePickerDialog.OnTimeSetListener mTimeSetListener = new
TimePickerDialog.OnTimeSetListener() {
        @Override
        public void onTimeSet(TimePicker view, int hour, int minute) {
            TimeDisplay.setText("Time: " + hour + ":" + minute);
        }
    };
    TimePickerDialog timeDialog = new TimePickerDialog(this, mTimeSetListener,
mHour, mMinute, true);
```

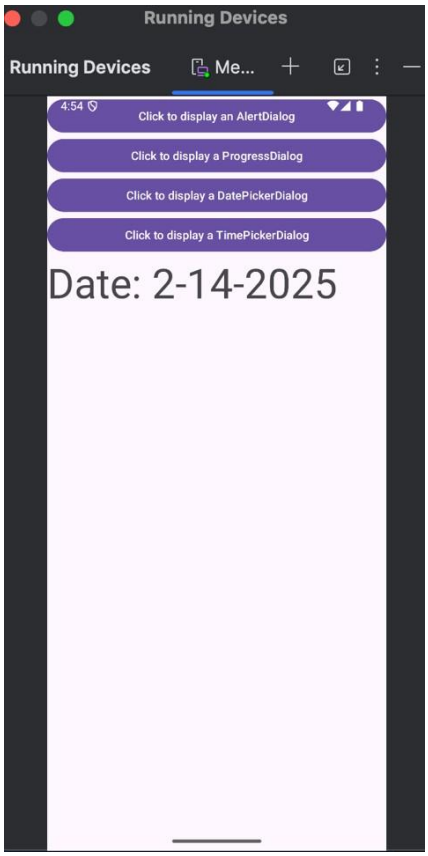
```
        timeDialog.show();
    }
}
```

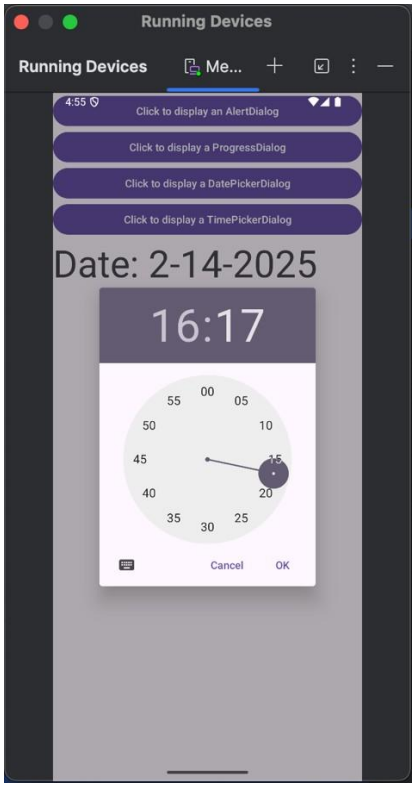
Output:

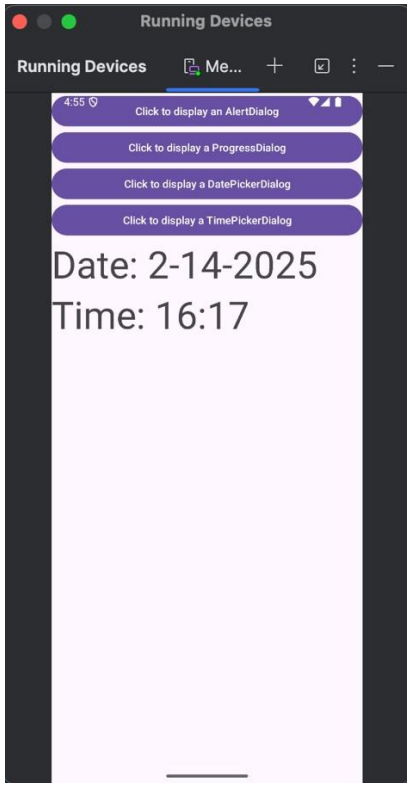





Running Devices
Running Devices
Me...
4:54
Click to display an AlertDialog
Click to display a ProgressDialog
Click to display a DatePickerDialog
Click to display a TimePickerDialog
2025
Fri, Feb 14
February 2025
S M T W T F S
Cancel
OK


Running Devices
Running Devices
Me...
4:54
Click to display an AlertDialog
Click to display a ProgressDialog
Click to display a DatePickerDialog
Click to display a TimePickerDialog
Date: 2-14-2025


Running Devices
Running Devices
Me...
4:55
Click to display an AlertDialog
Click to display a ProgressDialog
Click to display a DatePickerDialog
Click to display a TimePickerDialog
Date: 2-14-2025
16:17
Cancel
OK


Running Devices
Running Devices
Me...
4:55
Click to display an AlertDialog
Click to display a ProgressDialog
Click to display a DatePickerDialog
Click to display a TimePickerDialog
Date: 2-14-2025
Time: 16:17

Part C: -

Aim: Use Dialog fragments

Files Used:

Project Name: Prac_5c

Java File: MainActivity.java

XML File: activity_main.xml

Fragment File: BlankFragment.java & fragment_blank.xml

Java Class File: AlertDFragment.java

Steps to make a Fragment File same as 'Practical 2'

Code: fragment_blank.xml

```
<?xml version="1.0" encoding="utf-8"?>
<FrameLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".BlankFragment">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:gravity="center"
        android:text="This is a Dialog Fragment"
        android:textSize="20sp"/>
</FrameLayout>
```

BlankFragment.java

```
package com.example.prac_5c;

import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import androidx.fragment.app.DialogFragment;

public class BlankFragment extends DialogFragment {

    public BlankFragment() {
        // Required empty public constructor
    }

    @Override
    public View onCreateView(LayoutInflater inflater, ViewGroup container,
                             Bundle savedInstanceState) {
        // Inflate the layout for this fragment
        View rootView = inflater.inflate(R.layout.fragment_blank, container, false);
        getDialog().setTitle("Dialog Fragment");
        return rootView;
    }
}
```

AlertDFragment.java

```
package com.example.prac_5c;
import android.app.AlertDialog;
import android.app.Dialog;
import android.content.DialogInterface;
import android.os.Bundle;
import android.widget.Toast;
import androidx.annotation.NonNull;
import androidx.fragment.app.DialogFragment;

public class AlertDFragment extends DialogFragment {
    @Override
    public Dialog onCreateDialog(Bundle savedInstanceState) {
        return new AlertDialog.Builder(getActivity())
                .setIcon(android.R.mipmap.sym_def_app_icon)
                .setTitle("Alert Dialog Fragment")
                .setMessage("This is an Alert Dialog Fragment")
                .setPositiveButton("OK", new DialogInterface.OnClickListener() {
                    @Override
                    public void onClick(DialogInterface dialog, int which) {
                        Toast.makeText(getContext(), "OK Clicked",
Toast.LENGTH_SHORT).show();
                    }
                })
                .setNegativeButton("Cancel", new DialogInterface.OnClickListener() {
                    @Override
                    public void onClick(DialogInterface dialog, int which) {
                        Toast.makeText(getContext(), "Cancel Clicked",
Toast.LENGTH_SHORT).show();
                    }
                }).create();

            }}
```

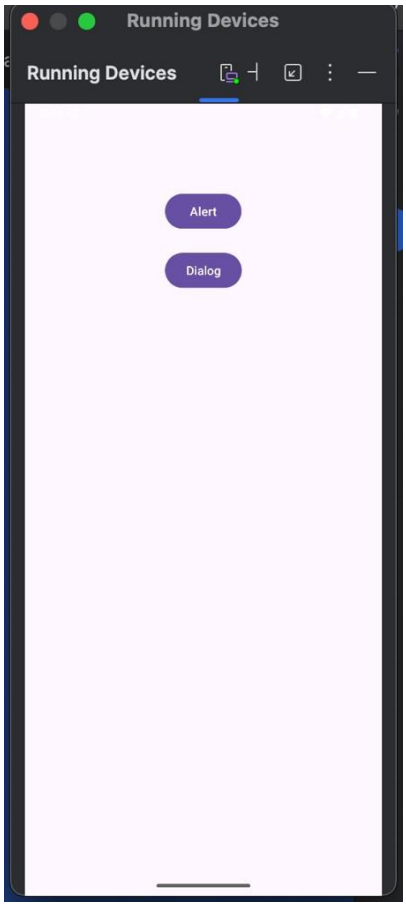
activity_main.xml

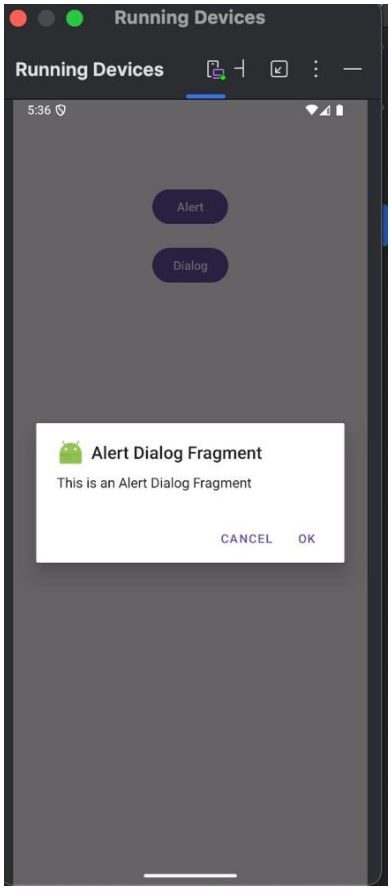
```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <Button
        android:id="@+id/button_alert"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Alert"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="100dp"
        android:onClick="showAlertDialog"/>

    <Button
        android:id="@+id/button_dialog"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Dialog"
        android:layout_below="@id/button_alert"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="20dp"
        android:onClick="showDialogFragment"/>
</RelativeLayout>
```

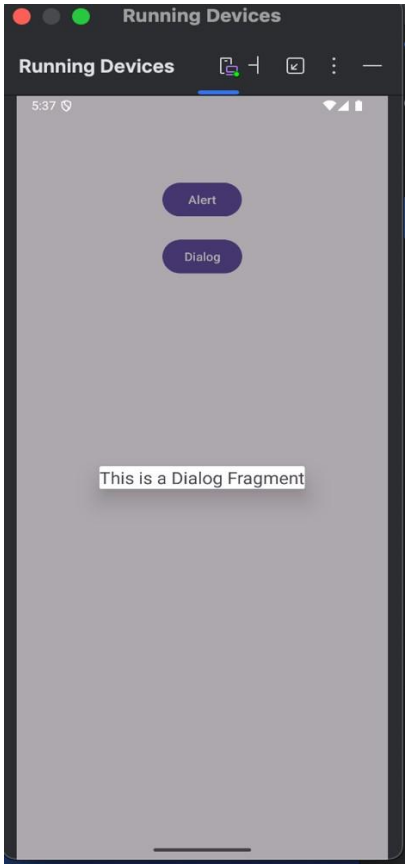
MainActivity.java

```
package com.example.prac_5c;

import android.os.Bundle;
import android.view.View;
import androidx.appcompat.app.AppCompatActivity;
import androidx.fragment.app.FragmentManager;

public class MainActivity extends AppCompatActivity {
    FragmentManager fm;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        fm = getSupportFragmentManager();
    }

    public void showDialogFragment(View view) {
        BlankFragment blankFragment = new BlankFragment();
        blankFragment.show(fm, "Dialog Fragment");
    }

    public void showAlertDialog(View view) {
        AlertDFragment alertDFragment = new AlertDFragment();
        alertDFragment.show(fm, "Alert Dialog Fragment");
    }
}
```

Output:


Running Devices
Running Devices
Alert
Dialog


Running Devices
Running Devices
5:36
Alert
Dialog
Alert Dialog Fragment
This is an Alert Dialog Fragment
CANCEL
OK


Running Devices
Running Devices
5:37
Alert
Dialog
This is a Dialog Fragment

Practical 6

Perform Intents, Events, Listeners and Adapters

Part A: -

Aim: Use The Android Intent Class using Events and Event Listeners.

Files Used:

Project Name: Prac_6a
Java File 1: MainActivity.java
Java File 2: new_page.java
XML File 1: activity_main.xml
XML File 2: activity_new_page.xml

Code: activity_new_page.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".new_page">
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="New Text"
        android:id="@+id/textView_view"
```

```
        android:textSize="20dp" />
</RelativeLayout>
```

new_page.java

```
package com.example.prac_6;

import static android.content.Intent.getIntent;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.widget.TextView;

public class new_page extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_new_page);

        TextView tv1 = (TextView) findViewById(R.id.textView_view);
        String myvalue = getIntent().getExtras().getString("my key");
        tv1.setText("Value = " + myvalue);
    }
}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
```

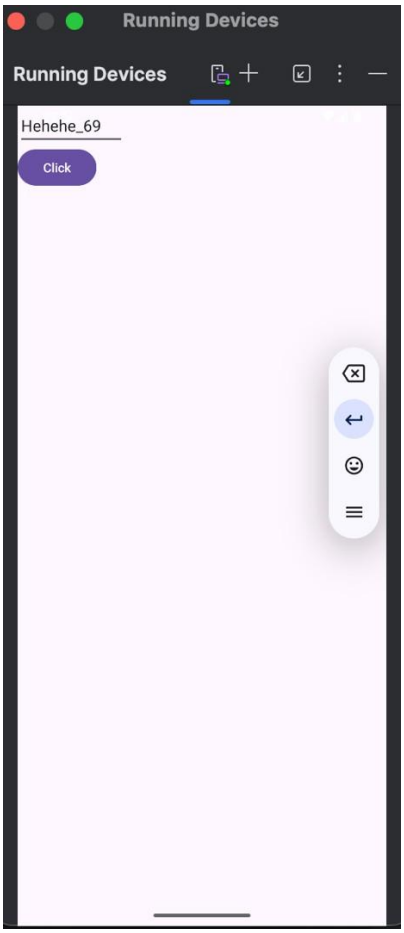
```
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    tools:context=".MainActivity">
    <EditText
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:hint="Enter a value :"
        android:id="@+id/editText_text" />
    <Button
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Click"
        android:id="@+id/button_click"
        android:layout_centerVertical="true"
        android:onClick="showntext"/>
</LinearLayout>
```

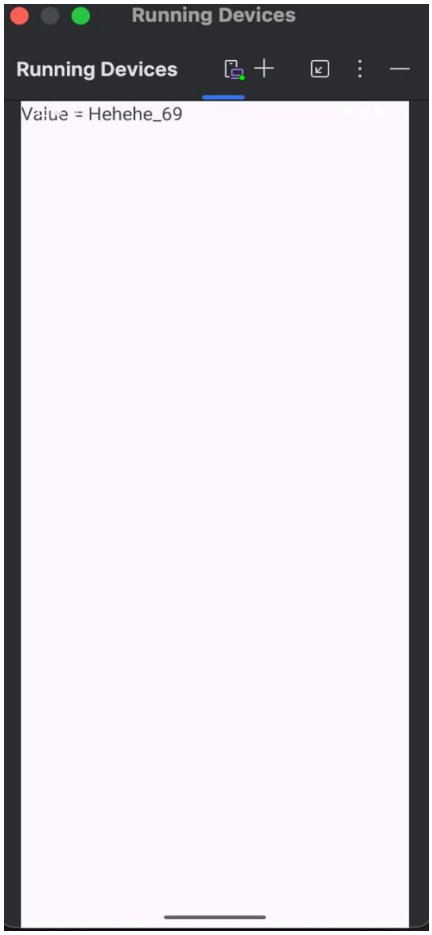
MainActivity.java

```
package com.example.prac_6;
import android.content.Intent;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.EditText;

public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
```

```
        setContentView(R.layout.activity_main);
    }
    public void showntext(View view) {
        EditText ed = (EditText) findViewById(R.id.editText_text);
        String msg = ed.getText().toString();
        Intent in = new Intent(this, new_page.class);
        in.putExtra("my key", msg);
        startActivity(in);
    }
}
```

Output:





Practical 7

Programs on Services, notification and broadcast receivers.

Part A: -

Aim: Use Services.

Files Used:

Project Name: Prac_7a
Java File: MainActivity.java
XML File: activity_main.xml
Java Class File: MyServices.java
AndroidManifest File: AndoridManifest.xml

Code: AndoridManifest.xml

```
<service android:name=".MyServices" android:exported="false"/>
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
```

```
    tools:context=".MainActivity">

    <Button
        android:id="@+id/btnStart"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Start Service"
        android:onClick="startService"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />

    <Button
        android:id="@+id/btnStop"
        android:onClick="stopService"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Stop Service"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@id/btnStart" />


</androidx.constraintlayout.widget.ConstraintLayout>
```

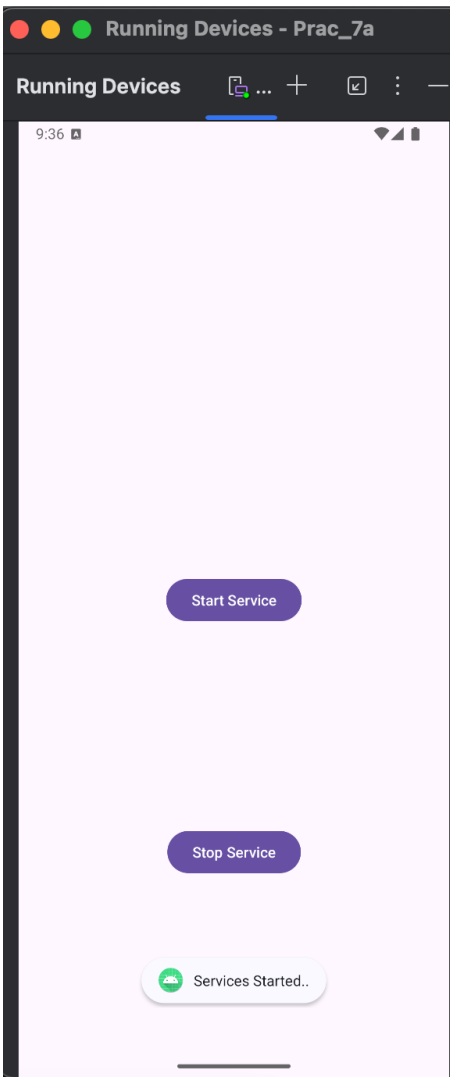
MainActivity.java

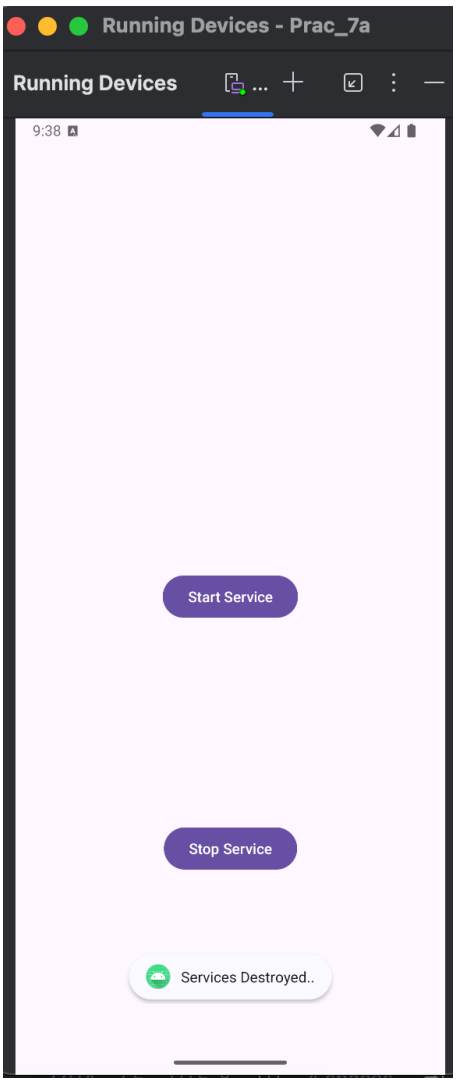
```
package com.example.prac_7a;

import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;


public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
    }
    public void startService(View view){
        Intent intent = new Intent(this,MyServices.class);
        startService(intent);
    }
    public void stopService(View view){
        Intent intent = new Intent(this,MyServices.class);
        stopService(intent);
    }
}
```

MyServices.java

```
package com.example.prac_7a;

import android.app.Service;
import android.content.Intent;
import android.os.IBinder;
import android.widget.Toast;

import androidx.annotation.Nullable;

public class MyServices extends Service {
    @Override
    public void onCreate() {
        super.onCreate();
    }

    @Override
    public int onStartCommand(Intent intent, int flags, int startId) {
        Toast.makeText(this, "Services Started..", Toast.LENGTH_LONG).show();
        return START_STICKY;
    }
    @Override
    public void onDestroy(){
        Toast.makeText(this,"Services Destroyed..",Toast.LENGTH_LONG).show();
    }

    @Nullable
    @Override
    public IBinder onBind(Intent intent) {
        return null;
    }
}
```

Output:


Running Devices - Prac_7a
Running Devices
9:36
Start Service
Stop Service
Services Started..


Running Devices - Prac_7a
Running Devices
9:38
Start Service
Stop Service
Services Destroyed..

Part B: -

Aim: Use Notification.

Files Used:

Project Name: Prac_7b
Java File: MainActivity.java
XML File: activity_main.xml
AndroidManifest File: AndoridManifest.xml

Code: AndoridManifest.xml

```
<uses-permission android:name="android.permission.POST_NOTIFICATIONS"/>
```

MainActivity.java

```
package com.example.prac_7b;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.NotificationCompat;
import androidx.core.app.NotificationManagerCompat;
import android.app.NotificationChannel;
import android.app.NotificationManager;
import android.os.Build;
import android.os.Bundle;
import android.widget.Toast;
import android.view.View;
```

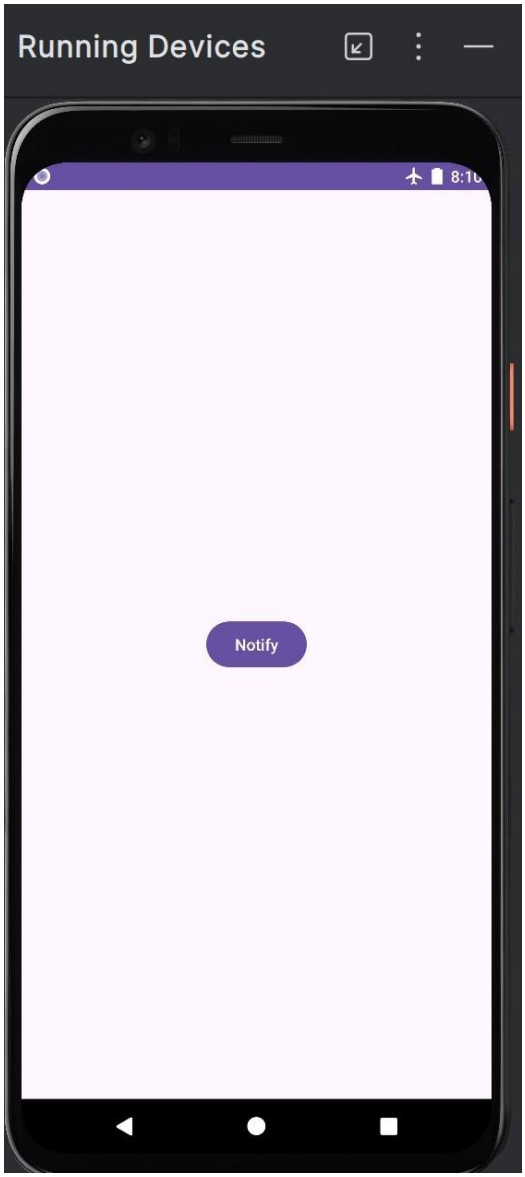
```
public class MainActivity extends AppCompatActivity {
public final String CHANNEL_ID="personal_notification";
public final int NOTIFICATION_ID=001;
   @Override
   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_main);
   }
   public void displayNotification(View view)
   {
      createNotificationChannel();
      Toast.makeText(getApplicationContext(), "Hi", Toast.LENGTH_LONG).show();
      NotificationCompat.Builder builder=new
NotificationCompat.Builder(this,CHANNEL_ID);
      builder.setSmallIcon(R.mipmap.ic_launcher);
      builder.setContentTitle("Simple Notification");
      builder.setContentText("THis is a test Notification");
      builder.setPriority(NotificationCompat.PRIORITY_DEFAULT);
      NotificationManagerCompat
notificationManagerCompat=NotificationManagerCompat.from(this);
      notificationManagerCompat.notify(NOTIFICATION_ID,builder.build());
   }
   private void createNotificationChannel()
   {
      if(Build.VERSION.SDK_INT>=Build.VERSION_CODES.O)
      {
         CharSequence name="Personal Notification";
         String description="This is description";
         int importance= NotificationManager.IMPORTANCE_DEFAULT;
          NotificationChannel notificationChannel=new
         NotificationChannel(CHANNEL_ID,name,importance);
```

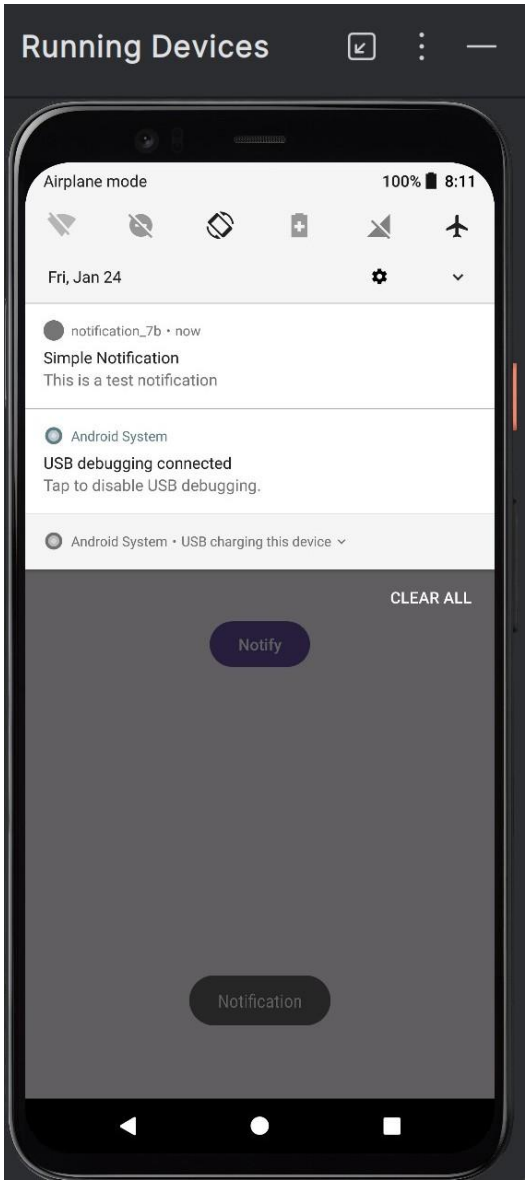
```
        notificationChannel.setDescription(description);
        NotificationManager
        notificationManager=(NotificationManager)getSystemService(NOTIFICATION_SE
        RVICE);
        notificationManager.createNotificationChannel(notificationChannel);
    }
  }
}
```

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
  xmlns:app="http://schemas.android.com/apk/res-auto"
  xmlns:tools="http://schemas.android.com/tools"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:orientation="vertical"
  tools:context=".MainActivity">
<Button
  android:layout_width="wrap_content"
  android:layout_height="wrap_content"
  android:layout_centerInParent="true"
  android:id="@+id/btn_noti"
  android:onClick="displayNotification"
  android:text="Notify"/>
</RelativeLayout>
```

Output:


Running Devices
Notify


Running Devices
Airplane mode
100% 8:11
Fri, Jan 24
notification_7b • now
Simple Notification
This is a test notification
Android System
USB debugging connected
Tap to disable USB debugging.
Android System • USB charging this device
CLEAR ALL
Notify
Notification

Part C: -

Aim: Use Broadcast Receiver.

Files Used:

Project Name: Prac_7c

Java File: MainActivity.java

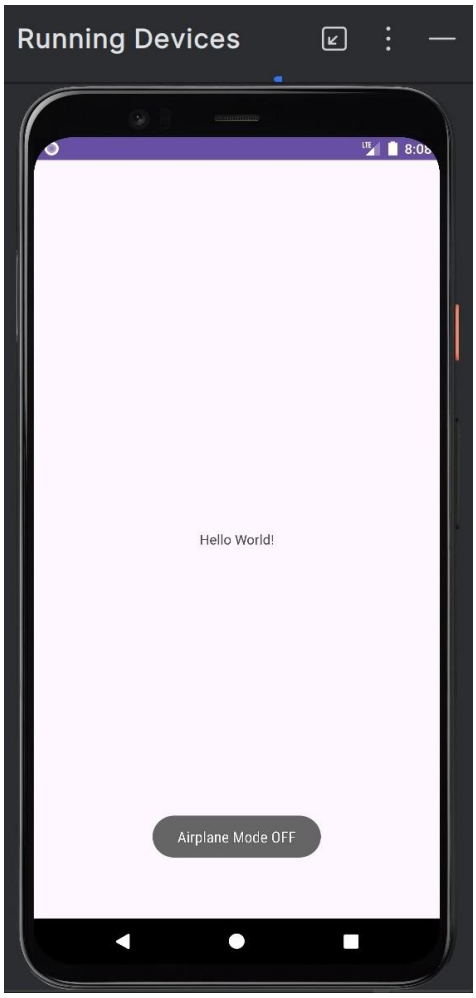
XML File: activity_main.xml

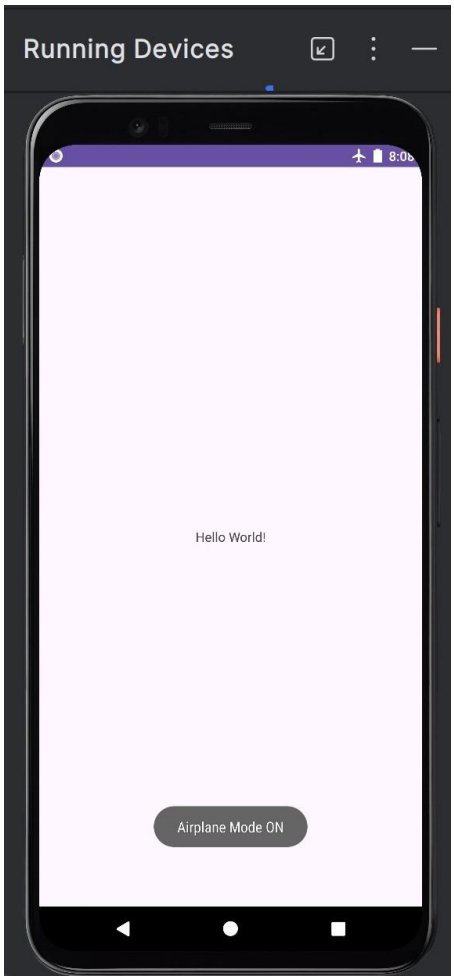
Code: MainActivity.java

```
package com.example.p7c;
import androidx.appcompat.app.AppCompatActivity;
import android.content.BroadcastReceiver;
import android.content.Context;
import android.content.Intent;
import android.content.IntentFilter;
import android.os.Bundle;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {

    private BroadcastReceiver airplaneModeReceiver = new BroadcastReceiver() {
        @Override
        public void onReceive(Context context, Intent intent) {
            boolean isAirplaneModeOn = intent.getBooleanExtra("state", false);
            Toast.makeText(context, isAirplaneModeOn ? "Airplane Mode ON" : "Airplane
Mode OFF", Toast.LENGTH_SHORT).show();
        }};
    @Override
```

```
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        registerReceiver(airplaneModeReceiver, new
IntentFilter(Intent.ACTION_AIRPLANE_MODE_CHANGED));
    }
    @Override
    protected void onDestroy() {
        super.onDestroy();
        unregisterReceiver(airplaneModeReceiver);
    }
}
```

Output:





Practical 8

Perform Database & JSON

Part A: -

Aim: Use Database Programming with SQLite

Files Used:

Project Name: Prac_8a

Java File: MainActivity.java

XML File: activity_main.xml

Code: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal"
        android:gravity="center_vertical"
        android:paddingBottom="8dp">

        <TextView
            android:id="@+id/user"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Username:" />


        <EditText
            android:id="@+id/etun"
```

```
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_weight="1"
        android:hint="Enter username"
        android:inputType="textPersonName"
        android:layout_marginStart="8dp" />
</LinearLayout>

<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal"
    android:gravity="center_vertical"
    android:paddingBottom="16dp">

    <TextView
        android:id="@+id/ipass"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Password:" />

    <EditText
        android:id="@+id/etp"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_weight="1"
        android:hint="Enter password"
        android:inputType="textPassword"
        android:layout_marginStart="8dp" />
</LinearLayout>

<Button
    android:id="@+id/reg"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Register" />

<Button
    android:id="@+id/lg"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="16dp"
    android:text="Login" />

<Button
    android:id="@+id/dr"
```

```
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="16dp"
        android:text="Drop" />

</LinearLayout>
```

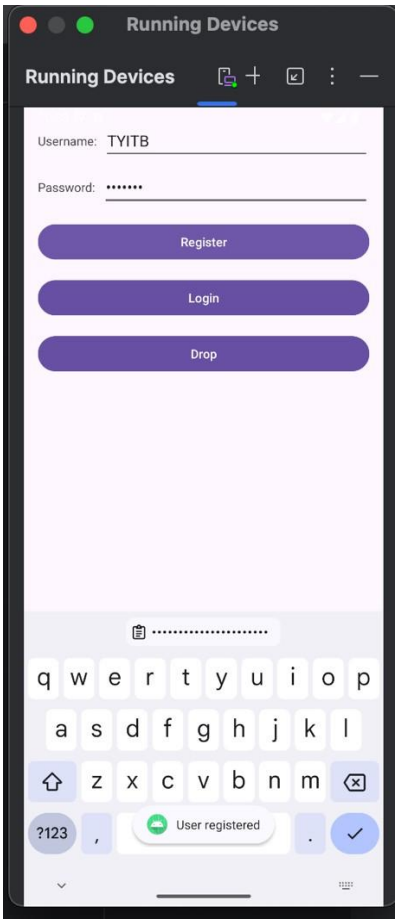
MainActivity.java

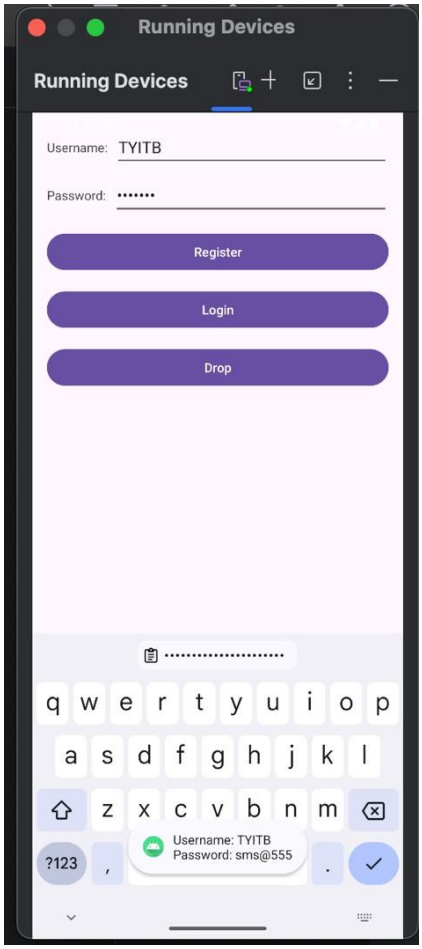
```
package com.example.prac_8a;

import androidx.appcompat.app.AppCompatActivity;
import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteException;
import android.database.sqlite.SQLiteOpenHelper;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

class MyDB extends SQLiteOpenHelper {
    MyDB(Context c) {
        super(c, "logindb", null, 1);
    }

    @Override
    public void onCreate(SQLiteDatabase db) {
        String str = "CREATE TABLE login(username TEXT, password TEXT)";
        db.execSQL(str);
    }

    @Override
    public void onUpgrade(SQLiteDatabase db, int oldVersion, int newVersion) {
        String dr = "DROP TABLE IF EXISTS login";
        db.execSQL(dr);
        onCreate(db);
    }
}

public class MainActivity extends AppCompatActivity {
```

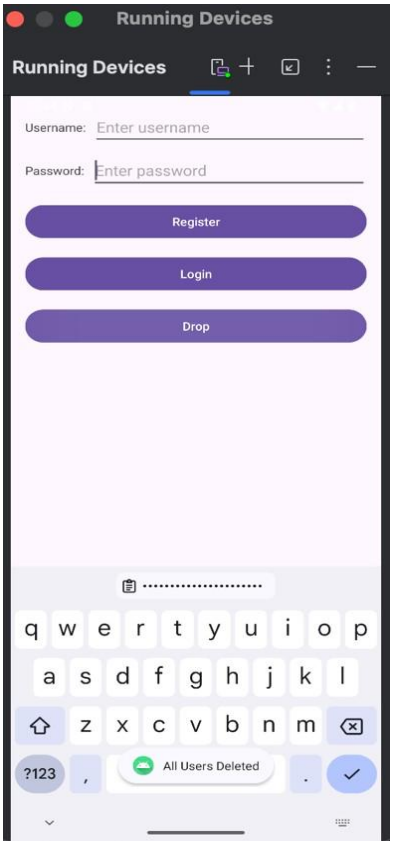
```
    MyDB mdb;
    EditText u, p;
    Button breg, bdr, blg;
    SQLiteDatabase db;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        u = findViewById(R.id.etun);
        p = findViewById(R.id.etp);
        blg = findViewById(R.id.lg);
        breg = findViewById(R.id.reg);
        bdr = findViewById(R.id.dr);

        mdb = new MyDB(this);

        // Set click listeners for buttons
        blg.setOnClickListener(this::login_fun);
        breg.setOnClickListener(this::reg_fun);
        bdr.setOnClickListener(this::dr_fun);
    }

    public void login_fun(View v) {
        String un = u.getText().toString();
        String pas = p.getText().toString();
        db = mdb.getReadableDatabase();

        String q = "SELECT * FROM login WHERE username=? AND password=?";
        try (Cursor c = db.rawQuery(q, new String[]{un, pas})) {
            if (c.getCount() == 0) {
                Toast.makeText(getApplicationContext(), "Username or password wrong or user
doesn't exist", Toast.LENGTH_SHORT).show();
            } else {
                while (c.moveToNext()) {
                    String uname = c.getString(0);
                    String passw = c.getString(1);
                    Toast.makeText(getApplicationContext(), "Username: " + uname + "\nPassword:
" + passw, Toast.LENGTH_LONG).show();
                    if (un.equals(uname) && pas.equals(passw)) {
                        Toast.makeText(getApplicationContext(), "Welcome user: " + un,
Toast.LENGTH_LONG).show();
                    }
                }
            }
```

```
        } catch (SQLiteException sqle) {
            sqle.printStackTrace();
        }
    }

    public void reg_fun(View v) {
        String un = u.getText().toString();
        String pas = p.getText().toString();
        db = mdb.getWritableDatabase();

        String q = "INSERT INTO login VALUES(?, ?)";
        db.execSQL(q, new Object[]{un, pas});
        Toast.makeText(getApplicationContext(), "User registered",
Toast.LENGTH_LONG).show();
    }

    public void dr_fun(View v) {
        u.setText("");
        p.setText("");
        db = mdb.getWritableDatabase();
        mdb.onUpgrade(db, 1, 2);
        Toast.makeText(getApplicationContext(), "All Users Deleted",
Toast.LENGTH_SHORT).show();
    }
}
```

Output:


Running Devices
Running Devices
Username: TYITB
Password: •••••••
Register
Login
Drop
User registered


Running Devices
Running Devices
Username: TYITB
Password: •••••••
Register
Login
Drop
Username: TYITB
Password: sms@555


Running Devices
Running Devices
Username: Enter username
Password: Enter password
Register
Login
Drop
All Users Deleted

Part B: -

Aim: Use Network Communications and Services (JSON).

Files Used:

Project Name: Prac_8b

Java File: MainActivity.java

XML File: activity_main.xml

AndroidManifest File: AndroidManifest.xml

build.gradle (:app)

Code: build.gradle (:app)

implementation"com.android.volley:volley:1.2.1"

NOTE: write this under dependencies.

AndroidManifest.xml

```
<uses-permission android:name="android.permission.INTERNET"/>
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    tools:context=".MainActivity">

    <Button
        android:id="@+id/button_fetch"
        android:layout_width="wrap_content"
```

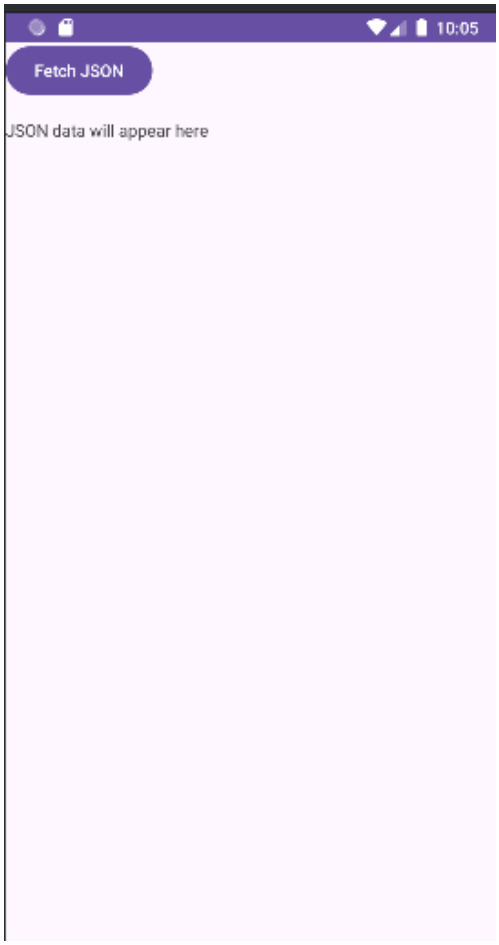
```
        android:layout_height="wrap_content"
        android:text="Fetch JSON"/>
    <TextView
        android:id="@+id/textView_result"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="16dp"
        android:text="JSON data will appear here"/>
</LinearLayout>
```

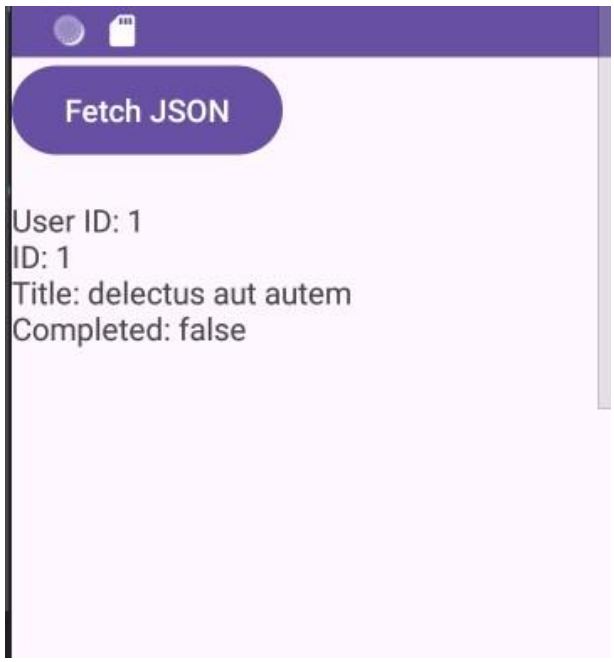
MainActivity.java

```
package com.example.prac_8b;

import androidx.appcompat.app.AppCompatActivity;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;

import com.android.volley.Request;
import com.android.volley.RequestQueue;
import com.android.volley.toolbox.Volley;
import com.android.volley.toolbox.JsonObjectRequest;
import com.android.volley.Response;
import com.android.volley.VolleyError;

import org.json.JSONObject;

public class MainActivity extends AppCompatActivity {
    TextView textViewResult;  // Declare TextView here for consistency

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        Button buttonFetch = findViewById(R.id.button_fetch);
        textViewResult = findViewById(R.id.textView_result);  // Assign the TextView
correctly

        buttonFetch.setOnClickListener(new View.OnClickListener() {
            @Override
```

```
            public void onClick(View v) {
                fetchJSON();
            }
        });
    }

    private void fetchJSON() {
        String url ="https://jsonplaceholder.typicode.com/todos/1";

        RequestQueue queue = Volley.newRequestQueue(this);

        JsonObjectRequest jsonObjectRequest = new JsonObjectRequest(
                Request.Method.GET, url, null,
                new Response.Listener<JSONObject>() {
                    @Override
                    public void onResponse(JSONObject response) {
                        try {
                            // Extracting all fields from the JSON response
                            String userId = response.getString("userId");
                            String id = response.getString("id");
                            String title = response.getString("title");
                            String completed = response.getString("completed");
                            // Displaying all the fields in the TextView
                            textViewResult.setText("User ID: " + userId + "\nID: " + id + "\nTitle: " +
title + "\nCompleted: " + completed);
                        } catch (Exception e) {
                            textViewResult.setText("Error: " + e.getMessage());
                        }
                    }
                },
                new Response.ErrorListener() {
                    @Override
                    public void onErrorResponse(VolleyError error) {
                        textViewResult.setText("Error: " + error.getMessage());
                    }
                }
        );
        queue.add(jsonObjectRequest);
    }
}
```

Output:


10:05
Fetch JSON
JSON data will appear here


Fetch JSON
User ID: 1
ID: 1
Title: delectus aut autem
Completed: false

Practical 9

Perform threads, handles and asynchronized programs.

Part A: -

Aim: Use Threads.

Files Used:

Project Name: Prac_9a

Java File: MainActivity.java

XML File: activity_main.xml

Code: activity_main.xml

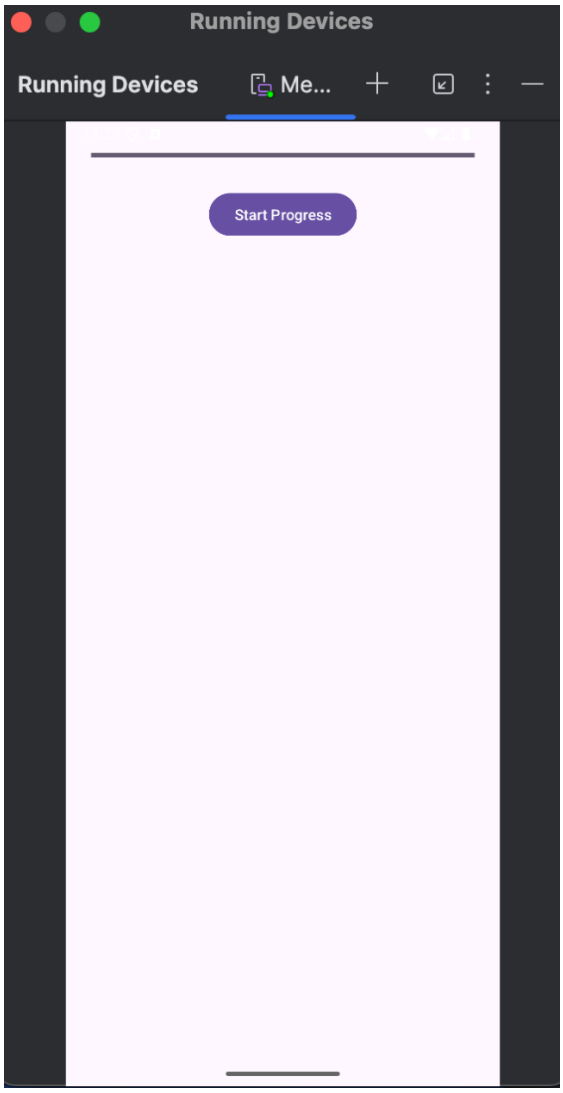
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp"
    tools:context=".MainActivity">
    <ProgressBar
        android:id="@+id/progressBar1"
        style="?android:attr/progressBarStyleHorizontal"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:indeterminate="false"
        android:max="10"
        android:padding="8dp" />
    <Button
        android:id="@+id/startButton"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_gravity="center_horizontal"
        android:text="Start Progress"
        android:onClick="startProgress"
```

```
        android:layout_marginTop="16dp" />
</LinearLayout>
```

MainActivity.java

```
package com.example.prac_9a;
import androidx.appcompat.app.AppCompatActivity;

import android.os.Bundle;
import android.view.View;
import android.widget.ProgressBar;

public class MainActivity extends AppCompatActivity {
    private ProgressBar bar;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        bar=(ProgressBar) findViewById(R.id.progressBar1);
    }
    public void startProgress(View view){
        bar.setProgress(0);
        new Thread(new Task()).start();
    }
    class Task implements Runnable{
        public void run(){
            for(int i=0;i<=10;i++)
            {
                final int value=i;
                try{
                    Thread.sleep(1000);
                }catch (InterruptedException e){
                    e.printStackTrace();
                }
                bar.setProgress(value);
            }
        }
    }
}
```

Output:



Part B: -

Aim: Use Handler.

Files Used:

Project Name: Prac_9b

Java File: MainActivity.java

XML File: activity_main.xml

Code: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp"
    tools:context=".MainActivity">

    <!-- Display Text Count -->
    <TextView
        android:id="@+id/textView"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Count"
        android:textSize="18sp"
        android:padding="8dp"
        android:layout_gravity="center_horizontal" />

    <Button
        android:id="@+id/bStartTimer"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Start Timer Count"
        android:onClick="timer"
        android:layout_marginTop="16dp" />
```

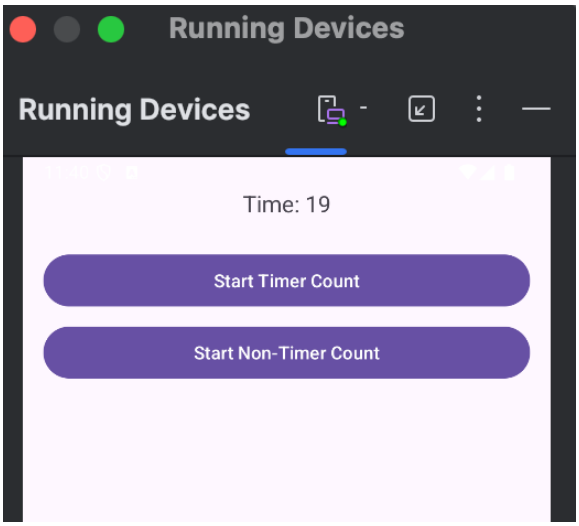
```
    <Button
        android:id="@+id/bStartNonTimer"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Start Non-Timer Count"
        android:onClick="nonTimer"
        android:layout_marginTop="8dp" />

</LinearLayout>
```

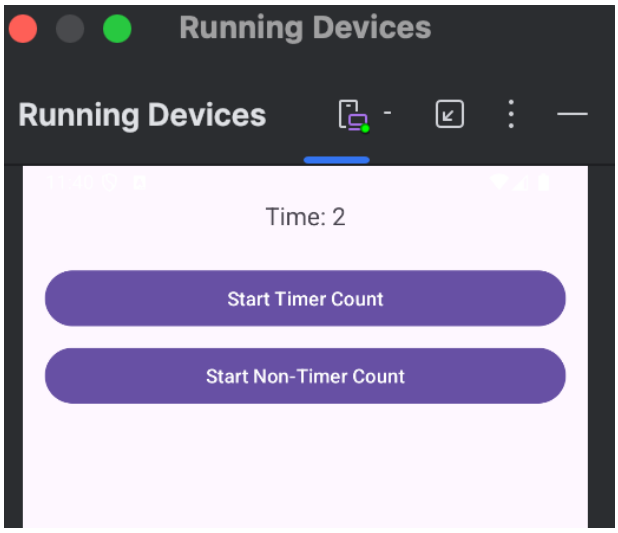
MainActivity.java

```
package com.example.prac_9b;

import android.os.Bundle;
import android.os.Handler;
import android.view.View;
import android.widget.ProgressBar;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    TextView textView;
    private int stopLoop = 30;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        textView = findViewById(R.id.textView);
    }

    public void timer(View view) {
        stopLoop = 30; // Reset timer value
        final Handler handler = new Handler();

        handler.post(new Runnable() {
            @Override
            public void run() {
                if (stopLoop > 0) {
                    stopLoop--;
                    textView.setText("Time: " + stopLoop);
                    handler.postDelayed(this, 1000); // Execute every 1 second
```

```
            }
        }
    });
}

public void nonTimer(View view) {
    // This method can freeze the app if it runs an infinite loop
    int i = 0;
    for (i = 0; i < 3; i++) {
        textView.setText("Time: " + i);
        try{
            Thread.sleep(1000);
        } catch (InterruptedException e){
            e.printStackTrace();
        }
    }
}
}
```

Output:





Part C: -

Aim: Use asynchronized.

Files Used:

Project Name: Prac_9c

Java File: MainActivity.java

XML File: activity_main.xml

Java Class File: MyTask.java

Code: MyTask.java

```
package com.example.prac_9c;

import android.app.ProgressDialog;
import android.content.Context;
import android.os.AsyncTask;
import android.widget.Button;
import android.widget.TextView;

public class MyTask extends AsyncTask<Void, Integer, String> {
    Context context;
    Button button;
    TextView textView;
    ProgressDialog progressDialog;

    MyTask(Context context, TextView textView, Button button) {
        this.context = context;
        this.textView = textView;
        this.button = button;
    }

    @Override
    protected void onPreExecute() {
        super.onPreExecute();
        progressDialog = new ProgressDialog(context);
        progressDialog.setTitle("Download in Progress");
        progressDialog.setMax(10);
        progressDialog.setProgress(0);
        progressDialog.setProgressStyle(ProgressDialog.STYLE_HORIZONTAL);
```

```
        progressDialog.show();
    }

    @Override
    protected void onPostExecute(String s) {
        super.onPostExecute(s);
        progressDialog.dismiss();  // Dismiss progress dialog after completion
        textView.setText("Download Finish");
        button.setEnabled(true);
    }

    @Override
    protected void onProgressUpdate(Integer... values) {
        super.onProgressUpdate(values);
        int progress = values[0];
        progressDialog.setProgress(progress);
        textView.setText("Download In Progress (" + progress + "/10)");
    }

    @Override
    protected String doInBackground(Void... voids) {
        for (int i = 1; i <= 10; i++) {
            try {
                Thread.sleep(1000); // Corrected the issue (previously used wait())
                publishProgress(i);
            } catch (InterruptedException e) {
                e.printStackTrace();
                return "Download Interrupted"; // Handle interruption properly
            }
        }
        return "Download Finish";
    }
}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
```

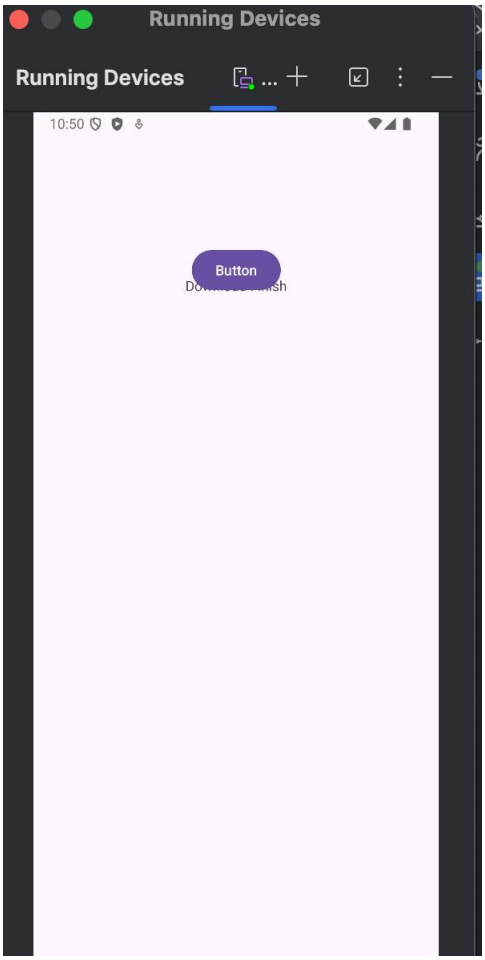
```
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/textView"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Hello World!"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintLeft_toLeftOf="parent"
        app:layout_constraintRight_toRightOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.168"/>
    <Button
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:id="@+id/button"
        android:layout_marginTop="112dp"
        android:text="Button"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"/>

</androidx.constraintlayout.widget.ConstraintLayout>
```

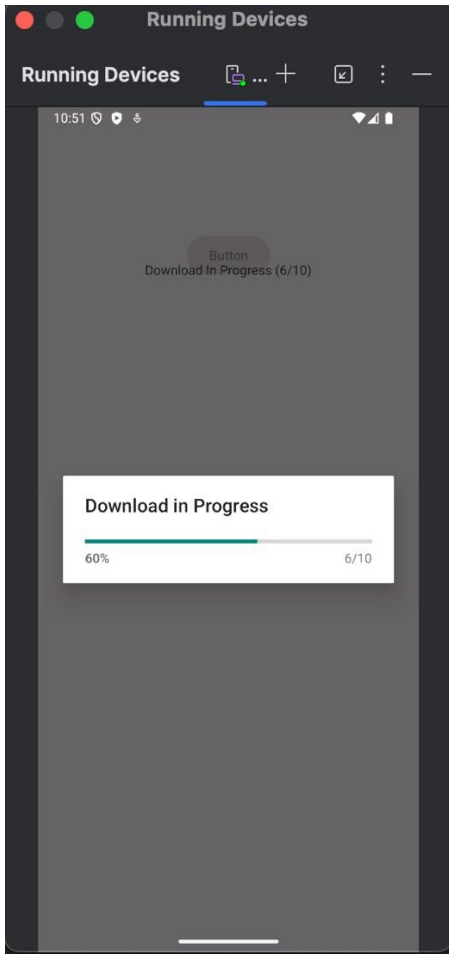
MainActivity.java

```
package com.example.prac_9c;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {
    TextView textView;
    Button button;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) ->
{
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });

        textView = findViewById(R.id.textView);
        button = findViewById(R.id.button); // Fixed the incorrect reference
```

```
        button.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                MyTask myTask = new MyTask(MainActivity.this, textView, button);
                myTask.execute();
                button.setEnabled(false);
            }
        });
    }
}
```

Output:





Practical 10

Perform Telecom APIs for media and telephone functions while ensuring security and proper permissions for safe access.

Part A: -

Aim: Use Media API.

Files Used:

Project Name: Prac_10a

Java File: MainActivity.java

XML File: activity_main.xml

Code: activity_main.xml

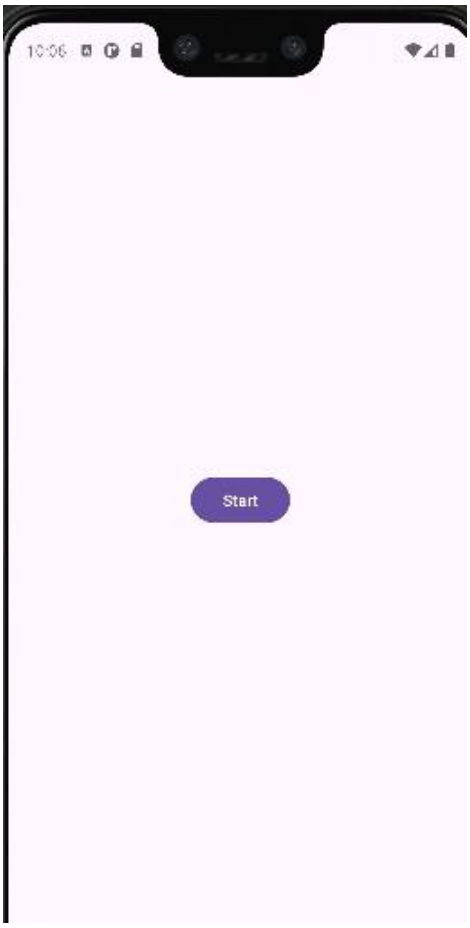
```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

   <!-- Button with id for reference -->
   <Button
       android:id="@+id/button"
       android:layout_width="wrap_content"
       android:layout_height="wrap_content"
       android:text="Start"
       android:layout_centerInParent="true"/>
</RelativeLayout>
```

MainActivity.java

```
package com.example.prac_10a;

import android.os.Bundle;
import android.media.MediaPlayer;
import android.view.View;
import android.widget.Button;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

    private Button bn;
    private MediaPlayer mediaPlayer;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        // Apply window insets for edge-to-edge display
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) ->
{
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
        // Initialize the button
        bn = findViewById(R.id.button);
        bn.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                mediaPlayer = MediaPlayer.create(getApplicationContext(), R.raw.song);
                mediaPlayer.start();
                bn.setEnabled(false);
                Toast.makeText(getApplicationContext(), "Playing music",
Toast.LENGTH_SHORT).show();
```

```
            // Set a completion listener for the media player
            mediaPlayer.setOnCompletionListener(new MediaPlayer.OnCompletionListener()
{
                @Override
                public void onCompletion(MediaPlayer mp) {
                    mediaPlayer.release();
                    mediaPlayer = null;
                    Toast.makeText(getApplicationContext(), "Playing Done",
Toast.LENGTH_SHORT).show();
                    bn.setEnabled(true);
                }
            });
        }
    });
  }
}
```

Output:



Part B: -

Aim: Telephone API.

Files Used:

Project Name: Prac_10b

Java File: MainActivity.java

XML File: activity_main.xml

Android Manifest File: AndroidManifest.xml

Code: AndroidManifest.xml

```
<uses-permission android:name="android.permission.CALL_PHONE"/>
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <EditText
        android:id="@+id/editText"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_alignParentTop="true"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="30dp" />

    <!-- Button to make call -->
    <Button
        android:id="@+id/button"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
```

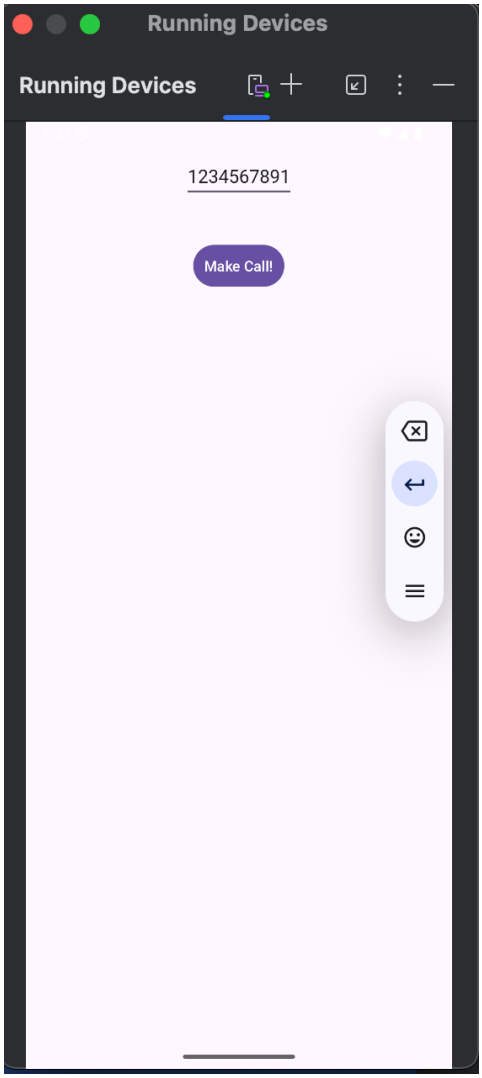
```
        android:layout_alignParentTop="true"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="115dp"
        android:padding="5dp"
        android:text="Make Call!" />
</RelativeLayout>
```

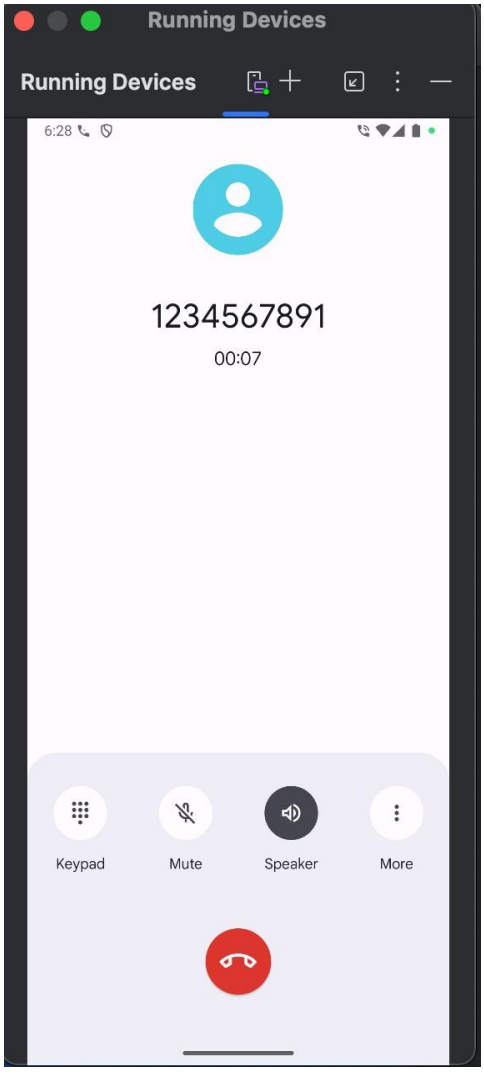
MainActivity.java

```
package com.example.prac_10b;

import android.Manifest;
import android.content.pm.PackageManager;
import android.net.Uri;
import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.ActivityCompat;
import androidx.core.content.ContextCompat;
import android.content.Intent;

public class MainActivity extends AppCompatActivity {
    EditText edittext;
    Button button;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        button = findViewById(R.id.button);
        edittext = findViewById(R.id.editText);

        button.setOnClickListener(view -> {
            String phone_number = edittext.getText().toString();

            // Check if CALL_PHONE permission is granted
            if (ContextCompat.checkSelfPermission(MainActivity.this,
Manifest.permission.CALL_PHONE)
                    != PackageManager.PERMISSION_GRANTED) {
                // Request permission
                ActivityCompat.requestPermissions(MainActivity.this,
```

```
                    new String[]{Manifest.permission.CALL_PHONE}, 1);
            } else {
                // Permission already granted, make the call
                makePhoneCall(phone_number);
            }
        });
    }

    private void makePhoneCall(String phone_number) {
        Intent phone_intent = new Intent(Intent.ACTION_CALL);
        phone_intent.setData(Uri.parse("tel:" + phone_number));
        startActivity(phone_intent);
    }

    @Override
    public void onRequestPermissionsResult(int requestCode, String[] permissions, int[]
grantResults) {
        super.onRequestPermissionsResult(requestCode, permissions, grantResults);
        if (requestCode == 1) {
            if (grantResults.length > 0 && grantResults[0] ==
PackageManager.PERMISSION_GRANTED) {
                // Permission granted, now make the call
                String phone_number = edittext.getText().toString();
                makePhoneCall(phone_number);
            } else {
                Toast.makeText(this, "Permission Denied!", Toast.LENGTH_SHORT).show();
            }
        }
    }
}
```

Output:


Running Devices
Running Devices
1234567891
Make Call!


Running Devices
Running Devices
6:28
1234567891
00:07
Keypad
Mute
Speaker
More

Part C: -

Aim: Use Security and permissions

Files Used:

Project Name: Prac_10c

Java File: MainActivity.java

XML File: activity_main.xml

AndroidManifest File: AndroidManifest.xml

Code: AndroidManifest.xml

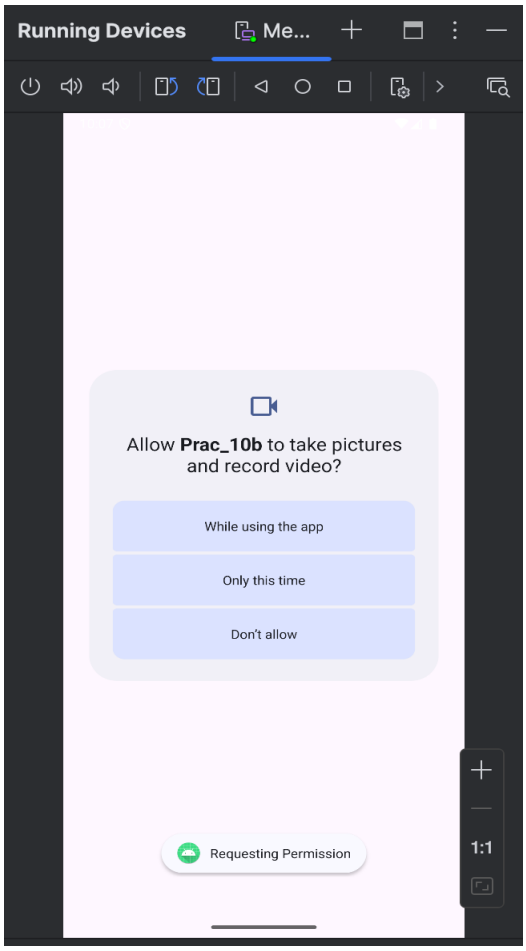
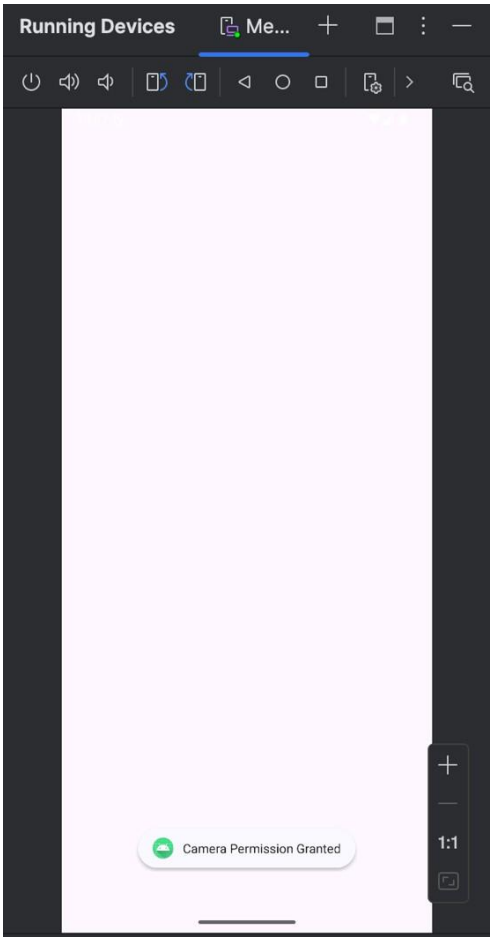
```
<uses-feature
    android:name="android.hardware.camera"
    android:required="false" />
<uses-permission android:name="android.permission.CAMERA" />
<uses-permission android:name="android.permission.ACCESS_FINE_LOCATION" />
<uses-permission android:name="android.permission.ACCESS_COARSE_LOCATION" />
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">
</RelativeLayout>
```

MainActivity.java

```
package com.example.prac_10c;

import android.content.pm.PackageManager;
import android.os.Bundle;
import android.widget.Toast;
import androidx.annotation.NonNull;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.ActivityCompat;
import androidx.core.content.ContextCompat;

public class MainActivity extends AppCompatActivity {
   private static final int CAMERA_PERMISSION_CODE = 100;

   @Override
   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_main);

      if (ContextCompat.checkSelfPermission(this, android.Manifest.permission.CAMERA)
            == PackageManager.PERMISSION_GRANTED) {
         // Permission is already granted
         Toast.makeText(this, "Permission is already granted",
Toast.LENGTH_SHORT).show();
      } else {
         Toast.makeText(this, "Requesting Permission", Toast.LENGTH_SHORT).show();
         requestCameraPermission();
      }
   }

   public void requestCameraPermission() {
      ActivityCompat.requestPermissions(this,
            new String[]{android.Manifest.permission.CAMERA},
            CAMERA_PERMISSION_CODE);
   }

   @Override
   public void onRequestPermissionsResult(int requestCode, @NonNull String[] permissions,
@NonNull int[] grantResults) {
      super.onRequestPermissionsResult(requestCode, permissions, grantResults);

      if (requestCode == CAMERA_PERMISSION_CODE) {
         if (grantResults.length > 0 && grantResults[0] ==
PackageManager.PERMISSION_GRANTED) {
```

```
            Toast.makeText(this, "Camera Permission Granted",
Toast.LENGTH_SHORT).show();
        } else {
            Toast.makeText(this, "Camera Permission Denied",
Toast.LENGTH_SHORT).show();
        }
    }
  }
}
```

Output: